Differentiated Policy Communication with Heterogeneous Agents*

Liangyuan Chen† Nannan Chen‡ Jiacheng Fan§ Jianhao Lin¶

Abstract

How should central banks design the precision of public signals for heterogeneous audiences? We study a Keynesian beauty-contest game in which a central bank communicates with experts and the general public through either a single signal or differentiated signals. The two groups differ in private-signal precision, interpretation costs, and coordination motives. Under a single signal, welfare is non-monotone in precision: higher precision sharpens inference about fundamentals but raises the public's interpretation cost and crowds out private information, so the optimal precision is strictly interior. Under differentiated signals, the central bank improves on the single-signal optimum by issuing a high-precision signal to experts and a simpler, low-precision signal to the general public. Differentiated communication works because it preserves expert-side accuracy while lowering the public's interpretation cost. The analysis rationalizes the recent central-bank practice of pairing technical policy statements with plain-language public summaries.

Keywords: policy communication, heterogeneous agents, information cost, beauty contest

JEL Classification: D82; E52; E58

*We gratefully acknowledge financial support from the Major Program of the National Social Science Foundation of China (Grant No. 24ZDA042) and the National Natural Science Foundation of China (Grant No. 72303258, 72273156, 723B2024). Any errors are our own.

†International School of Business and Finance, Sun Yat-sen University. E-mail: chenlyuan@mail.sysu.edu.cn.

‡School of Economics, Peking University. E-mail: 13622230562@163.com.

§Lingnan College, Sun Yat-sen University. E-mail: fanjch7@mail2.sysu.edu.cn.

¶Lingnan College, Sun Yat-sen University. E-mail: linjh3@mail.sysu.edu.cn. Corresponding author.

1 Introduction

Monetary policy today is as much about managing expectations as it is about setting interest rates. Woodford (2005) frames modern monetary policy as the art of expectation management: central banks shape beliefs through communication and thereby improve policy effectiveness (Blinder, 2009; Hansen et al., 2019). Recent policy discussions stress that communication must reach the broader public rather than financial markets alone (Blinder, 2018). Communication activity has expanded sharply over the past two decades (Haldane, 2017), yet the remaining difficulty is comprehension. Coenen et al. (2017) show that understanding European Central Bank statements requires thirteen to fifteen years of formal education, and survey evidence finds households poorly informed about central banks (van der Cruijsen et al., 2015). Relative to professional investors and economists, the general public is less equipped to process technical central-bank language. The question is no longer whether central banks should communicate, but how they should choose signal precision when audiences differ in interpretive capacity.

Central banks have already begun to adopt differentiated communication formats. Since November 2017, the Bank of England has published a plain-language summary of its *Inflation Report*, distilling Monetary Policy Committee decisions into accessible prose and graphics. Haldane and McMahon (2018) report a statistically significant increase in user activity on the Bank's website after this change. Bholat et al. (2019) also show that everyday language and relatable examples improve comprehension of monetary policy content. Comparable efforts are visible at the European Central Bank and the Bank of Canada (Draghi, 2014; Kozicki and Vardy, 2017).

These gains come with a trade-off. Technical communication is precise but hard to read; simplified communication is easier to process but may lose policy nuance and weaken effectiveness (Istrefi, 2019). Recent empirical work shows that the effect of policy communication depends on who receives which message and how the message is framed (Coibion et al., 2022, 2023; D'Acunto et al., 2022; Hoffmann et al., 2022). Existing models largely abstract from this trade-off by assuming homogeneous processing ability. This paper asks how a central bank should jointly choose public-signal precision and differentiated communication when experts and the general public differ in private-signal precision, interpretation cost, and coordination motive, three dimensions of heterogeneity that standard models treat as identical.

We study this problem in a Keynesian beauty contest with two groups, experts and the general public. Both groups choose actions under strategic complementarity, but they differ along three dimensions. Experts receive more precise private information. The general public faces an interpretation cost that rises with the precision of public communication, whereas experts interpret public signals at zero cost. In addition, both groups care about coordinating with the expert average rather than only matching fundamentals, so

public communication affects welfare both through information and through equilibrium coordination.

In the single-signal regime, the central bank releases one public signal and the model delivers a unique linear equilibrium. Both groups put more weight on the public signal than Bayesian precision weights alone would imply, because the coordination motive makes expert actions a payoff-relevant object. The public does so even more strongly than experts because its private signal is less precise. Welfare therefore depends on a *precision effect*, a *cost effect*, and a *crowding-out effect*. Higher public precision improves inference about fundamentals, but it also raises the public's interpretation cost and can crowd out socially valuable private information. Their interaction makes welfare non-monotone in precision and implies that the optimal single public signal is generically interior rather than maximal.

We then compare this benchmark with a differentiated-signal regime in which the planner keeps a high-precision signal for experts and adds a lower-precision signal for the general public. The equilibrium remains unique and linear, but the role of the two public signals becomes sharper. Both groups still overweight public information in total, yet within the composite public signal they weight the high- and low-precision signals in proportion to their precisions. The difference between experts and the public therefore lies in total reliance on public information, not in how the two signals are ranked internally. This structure delivers the main welfare result. Holding the expert-facing signal at the optimal single-signal precision, adding a second signal with very low precision strictly raises welfare, whereas duplicating the high-precision signal lowers welfare. Differentiated communication gains from separating informational content from cognitive accessibility: the central bank preserves expert-side accuracy through the high-precision signal while using the simpler signal to reduce the public's interpretation cost. The optimal low-precision signal therefore lies strictly below the expert signal, and the optimal precision gap widens when the general public is less informed, the expert sector is larger, or simplification is cheaper.

Finally, we allow the general public to choose which signals to interpret. In this extension the public may rationally ignore the high-precision signal and read only the simpler one when the extra information in the high-precision signal is too small relative to its interpretation cost. The planner responds by raising the precision of the simpler signal above its benchmark dual-signal level, because that signal becomes the public's effective communication channel. Under endogenous selective interpretation, differentiated communication remains optimal: one signal preserves expert accuracy, while the other steers the general public toward a lower-cost source of policy information. Differentiated communication therefore dominates both precision-only and readability-only benchmarks.

Related literature. The first literature studies the welfare value of public information in beauty-contest environments. Morris and Shin (2002) show that public signals improve coordination but may crowd out welfare-improving private information. Related work on

imperfect common knowledge, higher-order expectations, and the social value of information develops this logic (Allen et al., 2006; Angeletos and Lian, 2018; Angeletos and Pavan, 2007; Bacchetta and van Wincoop, 2006; Woodford, 2003). Svensson (2006) questions when the crowding-out logic is quantitatively relevant, and Cornand and Heinemann (2008) and Baeriswyl and Cornand (2014) show that optimal policy may involve limiting publicity across audiences. A recent set of heterogeneous-audience models allows some agents to be differently informed or to coordinate within groups (Chan, 2021; Grout et al., 2015). These papers explain why full transparency need not be optimal, but they are less suited to address a central feature of modern central bank communication: experts and the general public observe the same announcement yet respond to it differently because they differ not only in access, but also in private information, interpretation cost, and coordination motive.

A second body of work examines central bank communication with households and other non-experts. Households know little about central banks (van der Cruijsen et al., 2015), communication aimed at non-experts can be limited in effectiveness (Ehrmann and Wabitsch, 2022), and randomized information treatments show that the form of the message matters for household expectations (Coibion et al., 2022, 2023; D'Acunto et al., 2022; Hoffmann et al., 2022). Evidence comparing experts with representative samples further shows that the two groups often rely on systematically different subjective macroeconomic models (Andre et al., 2022). These papers document heterogeneous comprehension and responses, but they do not characterize the planner's optimal choice over signal precision and communication differentiation.

A third area studies multiple disclosures, attention allocation, information rigidities, and information costs. Morris et al. (2006), Morris and Shin (2007), and Myatt and Wallace (2014) show that the composition of public signals matters for welfare, and Baeriswyl et al. (2021) studies multiple signals without heterogeneous interpretation costs. Our setup shares the two-signal structure of Myatt and Wallace (2014) and Baeriswyl et al. (2021), but departs from them in three respects: (i) private-signal precision is heterogeneous across groups, (ii) interpretation costs enter asymmetrically, and (iii) coordination is tilted toward the expert average. These three features are what make the differentiated policy welfare-improving in our framework. In parallel, Arato et al. (2021) and Arato and Nakamura (2023) show that costly attention changes optimal transparency, and Coibion and Gorodnichenko (2015) shows that expectations adjust sluggishly when information diffuses slowly. Classic models emphasize the cost of acquiring information (Colombo and Femminis, 2014; Grossman and Stiglitz, 1980; Ui, 2014; Verrecchia, 1982). Our paper focuses instead on the cost of interpreting public information once it is available and asks how a central bank should choose signal precision and differentiated communication when experts and the general public coexist. This combination of features explains why differentiated communication can dominate a uniform disclosure and remain optimal when

the general public selectively interprets only the simpler signal.

The rest of the paper is organized as follows. Section 2 sets up the model and the equilibrium. Section 3 characterizes the optimal precision of a single public signal. Section 4 studies when differentiated communication improves welfare relative to the single-signal policy. Section 5 endogenizes the public's choice of which signals to interpret. Section 6 concludes.

2 Beauty-contest Model

2.1 Environment

The economy consists of two groups of agents with population shares λ (experts) and $1-\lambda$ (general public). Departing from the representative-agent frameworks of Morris and Shin (2002) and Baeriswyl et al. (2021), we allow the two groups to differ in private-signal precision, interpretation cost, and coordination motive. Experts are indexed by $i \in [0, \lambda]$ with actions $a_{i,e} \in \mathbb{R}$, and the general public is indexed by $i \in [\lambda, 1]$ with actions $a_{i,p} \in \mathbb{R}$.

Signals. All agents form beliefs about an unobservable economic fundamental θ. Each agent i of type $\tau \in \{e, p\}$ receives a private signal

$$x_{i,\tau} = \theta + \nu_{i,\tau}, \qquad \nu_{i,\tau} \sim \mathcal{N}\Big(0,\ \tfrac{1}{\beta_\tau}\Big), \tag{1}$$

where $\beta_\tau > 0$ is the precision of the private signal for type τ. Following Morris and Shin (2002), we place a diffuse prior on θ, so posterior beliefs are determined solely by the observed signals. Because experts possess superior professional knowledge and information-gathering ability, their private signals are more precise, so $\beta_e > \beta_p$. We normalize $\beta_e = 1$ throughout; the parameter $\beta \equiv \beta_p \in (0, 1)$ then measures the public's relative informational disadvantage.

In the baseline scenario the central bank releases a single public signal

$$y = \theta + \eta, \qquad \eta \sim \mathcal{N}\big(0,\ \tfrac{1}{\alpha}\big), \tag{2}$$

where $\alpha > 0$ is the precision of the public signal and the central bank's choice variable.

Utility function. The utility of agent i of type τ is

$$u_{i,\tau} = -(1-r)\big(a_{i,\tau} - \theta\big)^2 - r\big(L_{i,\tau} - \bar{L}\big) - C_\tau(\alpha), \tag{3}$$

where the three components capture distinct economic forces.

Fundamental loss. The first term, $-(1-r)(a_{i,\tau}-\theta)^2$, penalizes deviation of the agent's action from the true fundamental. This is the standard payoff-relevant motive: an agent

who aligns her action with θ avoids this loss entirely.

Coordination (beauty-contest) term. Motivated by the observation that market participants treat expert forecasters as reference points for their own expectations (Boun My et al., 2021), we redefine the coordination target so that every agent seeks to align with the *expert* average rather than with the entire population. Let $L_{i,\tau}$ denote agent i's mean-squared deviation from the expert average

$$L_{i,\tau} = \frac{1}{\lambda} \int_0^{\lambda} \left(a_{i,\tau} - a_{j,e}\right)^2 dj, \tag{4}$$

and $\bar{L}$ is the cross-sectional population average

$$\bar{L} = \int_0^{\lambda} L_{j,e}\, dj + \int_{\lambda}^{1} L_{j,p}\, dj \tag{5}$$

where the first term averages over expert actions and the second over the general public. The second term in (3), $-r(L_{i,\tau} - \bar{L})$, rewards any agent who is *closer* to the expert average than her peers. The parameter $r \in (0, 1)$ governs the strength of this coordination motive: a larger r indicates that agents attach greater weight to matching expert actions relative to tracking fundamentals. Because each individual i has zero measure in $[0, 1]$, her own action has only a negligible effect on $\bar{L}$, so the first-order condition reduces to the standard expression used below. This reformulation contrasts with Morris and Shin (2002), where the beauty-contest term is $L_i = \int_0^1 (a_i - a_j)^2\, dj$, making every agent an equally desirable coordination target. In our setting the asymmetry is empirically motivated: the general public revises its expectations toward expert forecasts, not toward the average layperson (Boun My et al., 2021).

Interpretation cost. The third term, $C_\tau(\alpha)$, reflects the cognitive burden of decoding the public signal. Indjejikian (1991) shows that information is costly to interpret and that investors' interpretive ability influences disclosure decisions; Orhun (2019) also emphasizes costly information interpretation. Because experts possess the professional knowledge required to parse policy statements, they bear no interpretation cost: $C_e(\alpha) \equiv 0$. The general public, by contrast, faces a cost $C_p(\alpha)$ that is continuously differentiable, strictly increasing, and convex in α:

$$C_p'(\alpha) > 0, \qquad C_p''(\alpha) \geq 0, \qquad \text{for all } \alpha > 0. \tag{6}$$

The monotonicity reflects that higher-precision signals convey more nuanced information but are harder for the general public to interpret. Central bank communications, for example, can require more than thirteen years of formal education to comprehend (Coenen et al., 2017), and the readability of financial disclosures affects how non-expert audiences process information (Fullwood, 2016). For a tractable benchmark, and following Boun My et al. (2021), we set $C_p(\alpha) = q\,\alpha$, where $q > 0$ is the marginal interpretation cost per unit

of precision. A larger q indicates that complex signals are harder to process. The Federal Reserve's pursuit of multiple policy objectives (e.g., price stability, maximum employment, and financial stability) has made central bank communications increasingly complex over time (Powell, 2018), which supports the assumption that greater signal precision raises the cognitive burden on the public.

Timing. The model unfolds in the following sequence. (1) The central bank commits to signal precision α. (2) Nature draws θ. (3) Agents simultaneously receive their private signals $x_{i,\tau}$ and observe the public signal y. (4) Agents simultaneously choose actions $a_{i,\tau}$.

Scenarios compared. To assess the value of differentiated communication, we compare two scenarios. In the *single-signal* scenario (Section 3), the central bank releases one public signal y with precision α to maximize social welfare W_1. In the *differentiated-signal* scenario (Section 4), it additionally releases a lower-precision signal aimed at the general public, yielding welfare W_2. The comparison between W_1 and W_2 isolates the welfare value of differentiated communication under heterogeneity.

2.2 Single-Signal Equilibrium

The unique symmetric linear equilibrium is characterized in the following lemma.

Lemma 1 (Equilibrium use of information)**.** *When the central bank releases a single public signal y with precision α, the unique linear Bayesian Nash equilibrium strategies are:*

$$a^1_{i,e} = \frac{1-r}{\alpha+1-r}\, x_{i,e} + \frac{\alpha}{\alpha+1-r}\, y, \tag{7}$$

$$a^1_{i,p} = \frac{(1-r)(\alpha+1)\beta}{(\alpha+\beta)(\alpha+1-r)}\, x_{i,p} + \frac{[\alpha+1-r+r\beta]\,\alpha}{(\alpha+\beta)(\alpha+1-r)}\, y. \tag{8}$$

Proof sketch. Maximizing (3) pointwise, the optimal action satisfies

$$a_{i,\tau} = (1-r)\,\mathbb{E}_{i,\tau}(\theta) + r\,\mathbb{E}_{i,\tau}(\bar{a}_e), \qquad \bar{a}_e = \frac{1}{\lambda}\int_0^\lambda a_{i,e}\, di. \tag{9}$$

Expert strategy. Conjecture a linear strategy $a^1_{i,e} = (1-k_e)\, x_{i,e} + k_e\, y$. The expert-average then collapses to

$$\bar{a}^1_e = \frac{1}{\lambda}\int_0^\lambda a^1_{i,e}\, di = (1-k_e)\theta + k_e y, \tag{10}$$

because private noise averages out. Using Bayes' rule,

$$\mathbb{E}_{i,e}(\theta \mid x_{i,e}, y) = \frac{x_{i,e} + \alpha\, y}{\alpha+1}. \tag{11}$$

Substituting (10)–(11) into (9) and matching coefficients yields

$$k_e = \frac{\alpha}{\alpha + 1 - r}, \tag{12}$$

which gives (7).

Public strategy. The general public observes y and $x_{i,p}$ but not individual expert signals. They therefore form expectations about the expert average using the reduced form (10):

$$\mathbb{E}_{i,p}(\bar{a}_e) = \mathbb{E}_{i,p}\big[(1 - k_e)\theta + k_e y\big] = (1 - k_e)\,\mathbb{E}_{i,p}(\theta) + k_e y. \tag{13}$$

With the Bayesian posterior $\mathbb{E}_{i,p}(\theta \mid x_{i,p}, y) = (\alpha y + \beta x_{i,p})/(\alpha + \beta)$, substituting into (9) and simplifying yields

$$k_p = \left[(1 - r)\frac{\alpha + 1}{\alpha + \beta} + r\right] k_e, \tag{14}$$

and collecting terms gives (8). A full derivation is provided in Appendix A. □

Economic interpretation. When $r = 0$, agents are pure Bayesians: weights are proportional to signal precisions, with the public signal receiving weight $\alpha/(\alpha + 1)$ for experts and $\alpha/(\alpha + \beta)$ for the public. When $r > 0$ the public signal is *overweighted* relative to its precision, because agents anticipate that other agents also over-respond to y; aligning with the expert average therefore requires placing extra weight on the signal the experts collectively overweight. A larger r amplifies this distortion, causing actions to deviate further from the fundamental.

The general public also places *strictly more* weight on the public signal than experts do. Formally,

$$k_p - k_e = \frac{\alpha(1 - r)(1 - \beta)}{(\alpha + 1 - r)(\alpha + \beta)} > 0, \tag{15}$$

since we normalize expert precision to unity ($\beta_e = 1$) and the public precision $\beta \equiv \beta_p \in (0, 1)$. Because the general public's private signal is less precise, it leans more heavily on the relatively accurate public signal, and this effect is amplified when the public also mimics experts' own overweighting of y.

The public's equilibrium action (8) depends on the expert precision (normalized to unity), even though the general public never observes expert signals directly. This dependence arises because the public infers the expert average $\bar{a}_e$ from the public signal y together with their knowledge of the equilibrium weight k_e; through this channel, the expert information structure is embedded in the public's optimal response. The converse does not hold: the expert strategy (7) is independent of β, reflecting the asymmetric

nature of the coordination game in which experts have no incentive to align with the general public.

3 Single-Signal Communication

3.1 Optimal Single-Signal Precision

Social welfare W_1 is the population average of individual utilities (where $u_{i,\tau}$ is given in (3), and the term $(1-\lambda)C_p(\alpha)$ is the aggregate interpretation-cost share borne by the general public). The cross-sectional mean of the coordination term $\int_0^1 (L_i - \bar{L})\, di$ vanishes by construction, so only the fundamental-loss and interpretation-cost components remain. The subscript 1 denotes the single-signal regime.

$$\begin{aligned} W_1 &= \int_0^1 u_i \, di \\ &= -(1-r) \int_0^1 \left(a_i^1 - \theta\right)^2 di - (1-\lambda)\, C_p(\alpha) \\ &= -\lambda(1-r)\frac{\alpha + (1-r)^2}{\left[\alpha + 1 - r\right]^2} - (1-\lambda)(1-r)\frac{\alpha + r^2 k_e^2 \beta}{\alpha\,(\alpha+\beta)} - (1-\lambda)\, q\, \alpha. \end{aligned} \tag{16}$$

Following Colombo and Femminis (2008), the interpretation cost is set as a linear function of signal precision, $C_p(\alpha) = q\,\alpha$, where $q > 0$ is the per-unit cost borne by the general public; a larger q reflects weaker financial literacy.

Morris and Shin (2002) (henceforth MS) show that welfare rises monotonically with public-signal precision under homogeneous processing ability. In our setting, precision also raises interpretation costs for the general public. To see how this changes the planner's problem, consider two limits of the first derivative:

$$\lim_{\alpha\to\infty} \frac{\partial W_1}{\partial \alpha} = -(1-\lambda)\, q < 0, \qquad \left.\frac{\partial W_1}{\partial \alpha}\right|_{\alpha=0} < 0 \quad \text{for } r > \tfrac{1}{2}. \tag{17}$$

The first limit implies that at high precision levels the marginal gain is dominated by the constant marginal cost q, so $\alpha \to \infty$ is never optimal when $q > 0$. The second limit implies that when coordination motives are strong ($r > 1/2$), welfare also falls near $\alpha = 0$ because agents over-weight a very noisy public signal. Together, these bounds yield a non-monotone profile: welfare falls, then rises, then falls again, with an interior optimum α^*.

To understand why, it is instructive to decompose W_1 into three structural components:

$$W_1 = W_{\text{precision}} + W_{\text{crowd-out}} + W_{\text{cost}}, \tag{18}$$

where each term is defined as follows.

Improving-precision effect. $W_{\text{precision}}$ measures social welfare in the hypothetical benchmark without coordination motives and without interpretation costs. In that benchmark agents' weights are determined purely by signal precisions, and social welfare is

$$W_{\text{precision}} = -(1-r)\left[\lambda \frac{1}{\alpha+1} + (1-\lambda)\frac{1}{\alpha+\beta}\right]. \tag{19}$$

A more precise public signal reduces the mean-squared deviation of agents' estimates from the fundamental θ, providing an unambiguous welfare gain from this channel alone.

Information-cost effect. W_{cost} captures the welfare loss attributable to interpretation costs:

$$W_{\text{cost}} = -(1-\lambda)\, q\, \alpha. \tag{20}$$

Because the cost function is linear in α, the marginal cost of raising precision is constant and does not diminish, so this channel exerts relentless downward pressure on welfare for all $\alpha > 0$.

Crowding-out effect. $W_{\text{crowd-out}}$ measures the welfare loss arising from agents' strategic over-reliance on the public signal driven by coordination incentives:

$$W_{\text{crowd-out}} = -(1-r)\left[\lambda \frac{r^2\,\alpha}{\big[\alpha+1-r\big]^2(\alpha+1)} + (1-\lambda)\frac{r^2\beta\,\alpha}{\big[\alpha+1-r\big]^2(\alpha+\beta)}\right]. \tag{21}$$

Because the general public places strictly more weight on the public signal than experts ($\beta \equiv \beta_p < \beta_e = 1$, $k_p > k_e$ by (15)), the public's relative overreaction to the public signal is larger. At the same time, the per-capita crowding-out loss for experts exceeds that for the general public:

$$\frac{r^2\,\alpha}{\big[\alpha+1-r\big]^2(\alpha+1)} > \frac{r^2\beta\,\alpha}{\big[\alpha+1-r\big]^2(\alpha+\beta)}, \tag{22}$$

since $1/(\alpha+1) > \beta/(\alpha+\beta)$ whenever $\beta < 1$. Intuitively, experts forgo a more precise private signal when they overweight the public signal, so the per-unit distortion is costlier. In aggregate, however, the general public usually drives the population-weighted loss because it comprises most of the population ($1-\lambda \gg \lambda$), and its aggregate crowding-out contribution is larger in total. Introducing a heterogeneous public group therefore amplifies the crowding-out distortion relative to the homogeneous-agent setting of Morris and Shin (2002). The superposition of all three effects renders the social value of increasing public-signal precision non-monotone.

Lemma 2 (Shape of the marginal welfare function)**.** *Define* $F(\alpha) \equiv \partial W_1/\partial\alpha + (1-\lambda)q$*, the marginal welfare benefit net of the precision and crowding-out effects but excluding the constant cost term.*

(i) When $r \leq 1/2$*,* $F(\alpha) > 0$ *for all* $\alpha > 0$*: the precision and crowding-out effects together always favour higher precision.*

(ii) When $r > 1/2$*,* F *first decreases and then increases, possessing a unique zero* $\hat{\alpha} \in (0, 2(1-r))$*. For* $\alpha < \hat{\alpha}$ *the combined precision–crowding-out effect is negative; for* $\alpha > \hat{\alpha}$ *it is positive.*

Proof. See Appendix A. □

Lemma 2 implies that when coordination motives are strong ($r > 1/2$), a small increase in public-signal precision can initially lower welfare because agents coordinate too strongly on a still-noisy public signal. Once precision crosses the threshold $\hat{\alpha}$, the informational gain dominates. At higher precision levels, interpretation costs eventually reverse the welfare gain. Combined with the constant marginal cost q, this shape delivers a unique interior optimum α^*.

Assumption 1 (Strong Coordination)**.** *We focus on* $r > \frac{1}{2}$ *throughout. Under this condition, the coordination motive is strong enough that a very noisy public signal can lower welfare, generating a non-monotone welfare profile. The case* $r \leq \frac{1}{2}$ *yields a monotone profile and is discussed in Lemma 2(i).*

For the quantitative benchmark, we set $r = 0.85$, normalize expert private-signal precision to one, set $\beta = 0.4$, choose $\lambda = 0.005$, and set $q = 0.04$. The value of λ should be read as a calibration of a small expert audience rather than as a literal population share; robustness to $\lambda \in \{0.005, 0.02, 0.05\}$ leaves the qualitative shape of W_1 unchanged. Figure 1 plots W_1 against α for four model specifications: (i) the baseline of Morris and Shin (2002) (MS), (ii) heterogeneous agents without interpretation costs, (iii) interpretation costs without heterogeneity, and (iv) the full model. Welfare is maximized at a finite interior α^*, a substantial improvement over no communication (see Figure 1). The benchmark optimum lies above both β and the expert private precision, which is consistent with Svensson's (2006) argument that central bank forecasts should be more accurate than private-sector forecasts. Unlike the MS benchmark, however, welfare does not keep improving with precision: arbitrarily complex communication is not optimal once interpretation costs are taken seriously.

Assumption 2 (Moderate interpretation cost)**.** *The marginal interpretation cost* q *is below a threshold* $\bar{q} > 0$*, so that the interior optimum* α^* *satisfying* $\partial W_1/\partial\alpha = 0$ *yields a strict welfare improvement over no communication:* $W_1(\alpha^*) > W_1(0)$*.*

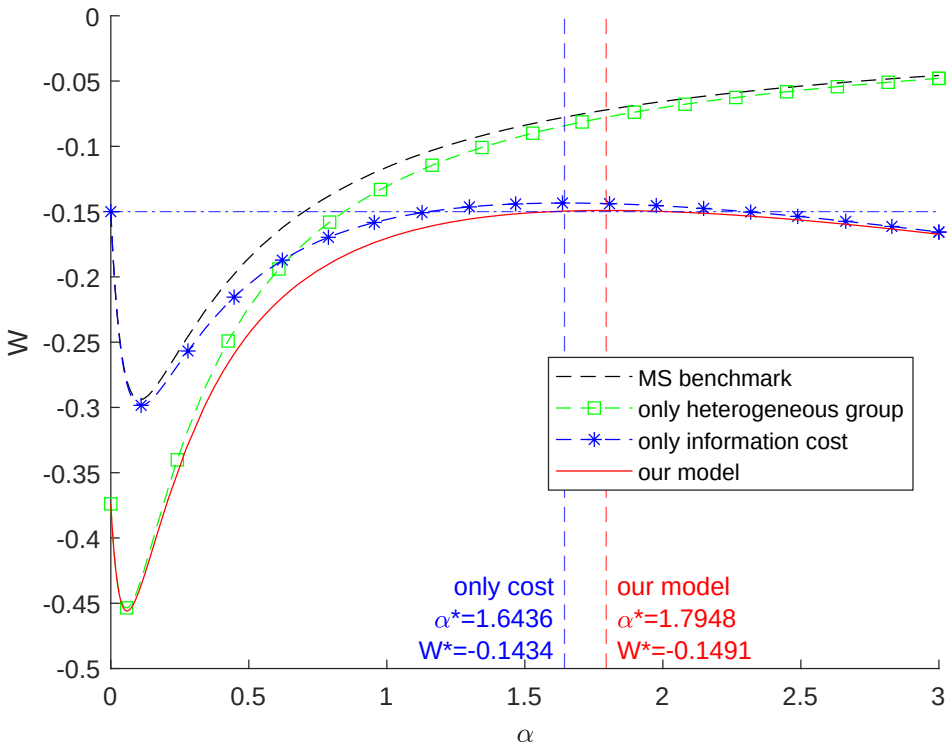


Figure 1: Social welfare and optimal precision of public information.
Note: "MS" denotes Morris and Shin (2002) with $r = 0.85$, $\beta = 1$. "Heterogeneous agents only" adds two agent types without interpretation cost. "Interpretation cost only" adds cost without heterogeneity. "Full model" incorporates both. Parameters: $r = 0.85$, $\beta = 0.4$, $\lambda = 0.005$, $q = 0.04$.

This assumption ensures that the general public's comprehension capacity is sufficient to make communication worthwhile. All subsequent results condition on $q < \bar{q}$. The threshold $\bar{q}$ depends on the remaining parameters (β, r, λ) and is positive whenever $\beta < 1$. Under the benchmark calibration, $\bar{q}$ is well above $q = 0.04$, so the baseline value lies well inside the region where communication raises welfare.

Proposition 1 (Interior optimum of public-signal precision)**.** *When the economy contains two types of heterogeneous agents and the general public bears a linear interpretation cost $C_p(\alpha) = q\alpha$ with $q > 0$, full transparency ($\alpha \to \infty$) is never socially optimal. There exists a finite precision $\alpha^* < \infty$ that maximizes social welfare W_1.*

Proof. See Appendix A. □

Proposition 1 follows from a simple comparison of marginal benefit and marginal cost. Sharper communication reduces uncertainty and, once precision is high enough, also mitigates the crowding-out distortion. But the marginal interpretation cost remains strictly positive. Because the derivative in equation (17) is negative for sufficiently large α, the welfare function must attain its maximum at a finite precision.

3.2 Comparative Statics of Optimal Precision

Relative to Morris and Shin (2002), our model adds population heterogeneity and interpretation costs. We now examine how these features shift the optimal precision α^*.

Public precision β. Holding $\lambda = 0.005$, $r = 0.85$, expert precision normalized to unity, and $q = 0.04$ fixed, Figure 2b shows that α^* is decreasing in β. When the public's own private information is already relatively precise, the social planner need not supply as much supplementary information through the public channel. In the limit $\beta \to 1$, the agents are nearly homogeneous and the optimal precision falls to the level that merely compensates for interpretation costs. Conversely, when β is very small, each extra unit of public precision yields a larger gain in the public's posterior accuracy, so the planner sets a higher α^*. For all $\beta \in (0, 1]$, α^* remains a finite real number, confirming that the social planner's optimal strategy is neither completely opaque nor completely transparent.

Expert share λ. Figure 2d shows that α^* is increasing in the proportion of experts λ. As the expert share grows, the population-average cost burden $(1 - \lambda)\, q\, \alpha$ shrinks, so the social planner can afford to release a more precise, and hence more complex, public signal. In the degenerate limit $\lambda = 1$ (all agents are experts), interpretation costs vanish entirely and the model collapses to Morris and Shin (2002), in which releasing information as precisely as possible is always optimal. This result provides a theoretical rationale for the "3 E's" (explanation, engagement, and education) advocated by Haldane et al. (2020) as a central bank communication strategy: by raising the general public's financial literacy, the central bank effectively increases the share of the population that can process complex information at low cost, which in turn permits a more informative communication policy.

Marginal interpretation cost q. Figure 2f shows that, under Assumption 2, α^* is strictly decreasing in q: as interpretation becomes easier, the planner can raise precision without incurring prohibitive costs. The threshold $\bar{q}$ formalises the common observation that central bank communication presupposes a minimum level of financial literacy on the part of the audience it addresses.

Proposition 2 (Comparative Statics of Optimal Precision)**.** *Under Assumption 2, the optimal public-signal precision α^* is (i) decreasing in the general public's private-signal precision β; (ii) increasing in the expert share λ; and (iii) decreasing in the marginal interpretation cost q.*

Proof. See Appendix A. □

The common thread across all three comparative-static results is that α^* rises whenever the social planner can extract more value from precision (either because the public lacks

private information, or because experts, who bear no interpretation cost, form a larger share of the population) or whenever the unit cost of extracting that value is lower.

4 Differentiated Communication

4.1 Dual-Signal Environment

When the economy contains agents with heterogeneous comprehension capacities, a single public signal may be simultaneously too complex for the general public and insufficiently nuanced for experts. A natural question is whether the social planner can do better by issuing two public signals that differ in precision and in linguistic accessibility. This approach is already visible in policy practice. Haldane (2017) shows, using Flesch–Kincaid readability scores, that traditional central bank publications require a reading grade of 14 to 18, roughly university level, which makes them inaccessible to at least 90% of the general public. In response, the Bank of England introduced a Visual Summary of its Inflation Report in November 2017, written in plain language and illustrated with graphics. The European Central Bank (Draghi, 2014) and the Bank of Canada (Kozicki and Vardy, 2017) have adopted analogous simplifications. Istrefi (2019) notes, however, that the simplified summary omits important information about outlook uncertainty and so gives readers a misleading sense of certainty. Whether the welfare gain from improved public comprehension outweighs the welfare loss from reduced precision remains an open question.

To study this trade-off formally, suppose the social planner releases two public signals:

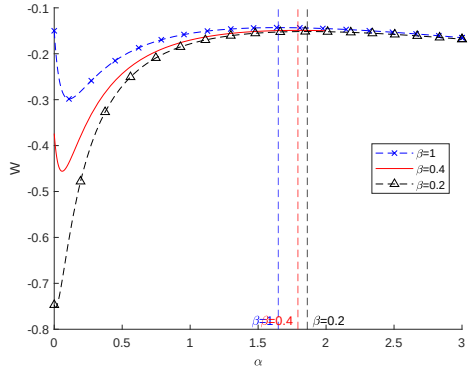
$$y^L = \theta + \eta^L, \quad \eta^L \sim N\left(0, \tfrac{1}{\alpha_L}\right); \qquad y^H = \theta + \eta^H, \quad \eta^H \sim N\left(0, \tfrac{1}{\alpha_H}\right). \tag{23}$$

Both signals y^L and y^H are publicly broadcast and observed by all agents, experts and general public alike. The distinction between the two signals lies not in who observes them but in the cost of interpretation: the general public bears cost q_L to process y^L and cost q_H to process y^H, whereas experts process both at zero cost.

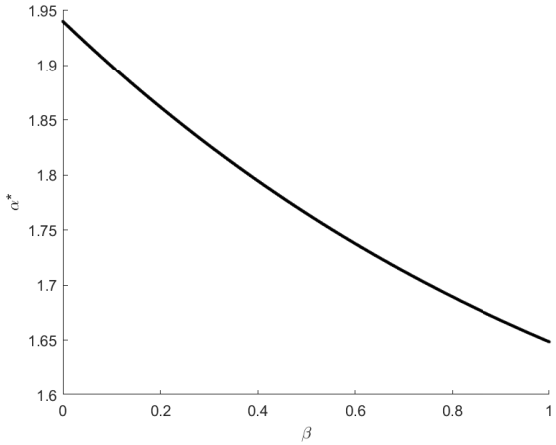
Here y^L is the low-precision signal, expressed in simple language and therefore easy to interpret, with precision α_L. Signal y^H is the high-precision signal with precision α_H, which is informative but demanding to comprehend.

Assumption 3 (Signal Ordering)**.** *The differentiated-signal regime proper corresponds to $\alpha_H > \alpha_L > 0$. The boundary cases $\alpha_L = 0$ and $\alpha_L = \alpha_H$ collapse the regime to a single effective signal and are used below only as comparison benchmarks.*

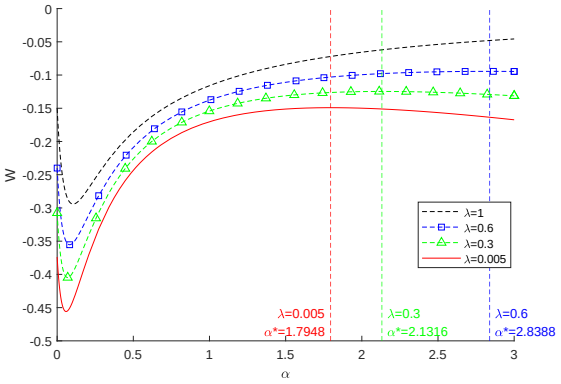
Following Baeriswyl and Cornand (2014) and Baeriswyl et al. (2021), the noise terms η^L and η^H are assumed uncorrelated; the case of correlated signals yields qualitatively identical conclusions and is treated in Appendix D. Independence is the natural baseline when y^L and y^H arise from separate communication processes, for example a technical



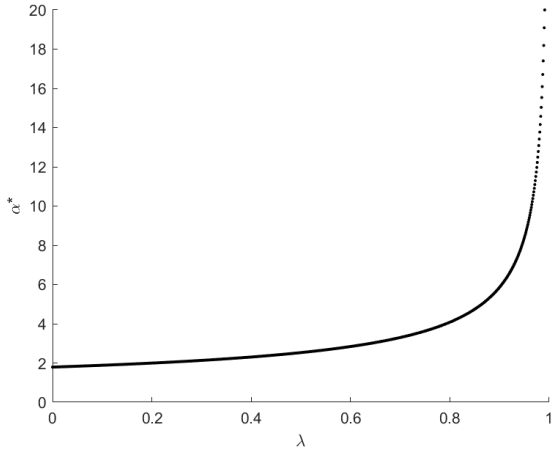
(a) W_1 under different β



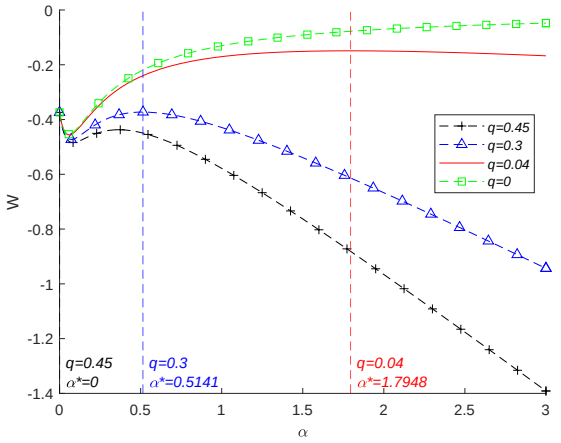
(b) α^* as a function of β



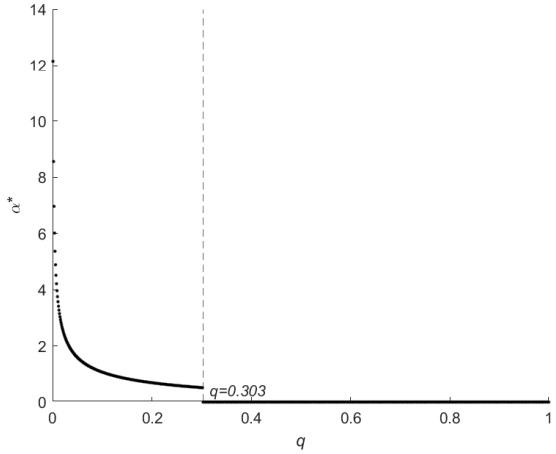
(c) W_1 under different λ



(d) α^* as a function of λ



(e) W_1 under different q



(f) α^* as a function of q

Figure 2: Comparative statics of the optimal single-signal precision α^*.
Note: Panels (a)–(b): sensitivity to public private-signal precision β. Panels (c)–(d): sensitivity to expert share λ. Panels (e)–(f): sensitivity to marginal interpretation cost q. Benchmark: $r = 0.85$, $\beta = 0.4$, $\lambda = 0.005$, $q = 0.04$.

policy report and a plain-language press release, rather than from two presentations of the same source material.

Interpretation costs in the differentiated-signal regime are adjusted as follows. Experts incur no cost interpreting either signal, $C_e = 0$. For the general public, the total cost is $C_p(\alpha_L, \alpha_H) = q_L\,\alpha_L + q_H\,\alpha_H$. Two restrictions govern the cost parameters. First, the low-quality signal is cheaper to interpret:

$$q_L\,\alpha_L < q_H\,\alpha_H. \tag{24}$$

Second, in the limiting case $\alpha_L = \alpha_H$ the two signals are identical and there is no meaningful quality distinction, so costs should coincide. These two requirements are jointly satisfied by setting

$$q_L = q_H \cdot \frac{\alpha_L}{\alpha_H}, \qquad q_H = q, \tag{25}$$

where q is the same unit-cost parameter as in the single-signal benchmark. This parameterization ensures $q_L \leq q_H$ with equality if and only if $\alpha_L = \alpha_H$. The proportional structure captures the intuition that the cognitive burden of processing a signal scales with its informational content: a signal of lower precision conveys less information and therefore imposes a smaller interpretation cost. Under parameterization (25), condition (24) reduces to $\alpha_L < \alpha_H$, which coincides with the signal-ordering condition; the two restrictions are therefore not independent.

4.2 Dual-Signal Equilibrium

In the welfare analysis (Section 4.3), we fix $\alpha_H = \alpha^*$ and optimize over α_L alone, so any welfare gain is attributable purely to the addition of the low-precision signal.

Lemma 3 (Dual-Signal Equilibrium)**.** *When the central bank releases y^L and y^H with precisions α_L and α_H, there exists a unique linear Bayesian Nash equilibrium. The expert's equilibrium action is*

$$a_{i,e}^{L,H} = \frac{1-r}{\alpha_L + \alpha_H + 1 - r}\,x_{i,e} + \frac{\alpha_L}{\alpha_L + \alpha_H + 1 - r}\,y^L + \frac{\alpha_H}{\alpha_L + \alpha_H + 1 - r}\,y^H. \tag{26}$$

The general public's equilibrium action is $a_{i,p}^{L,H} = (1 - k_{L,p} - k_{H,p})\,x_{i,p} + k_{L,p}\,y^L + k_{H,p}\,y^H$, with weights

$$k_{L,p} = \frac{\left[(1-r)\dfrac{\alpha_H + \alpha_L + 1}{\alpha_H + \alpha_L + \beta} + r\right]\alpha_L}{\alpha_H + \alpha_L + 1 - r}, \tag{27}$$

$$k_{H,p} = \frac{\left[(1-r)\dfrac{\alpha_H + \alpha_L + 1}{\alpha_H + \alpha_L + \beta} + r\right]\alpha_H}{\alpha_H + \alpha_L + 1 - r}. \tag{28}$$

Proof. See Appendix B. □

In the differentiated-signal regime every agent i observes three signals: private signal x_i, low-quality public signal y^L, and high-quality public signal y^H. The first-order condition for individual utility maximization yields the linear equilibrium action:

$$a_{i,\tau} = (1-r)\, E_{i,\tau}(\theta) + r\, E_{i,\tau}(\bar{a}_e), \qquad \bar{a}_e = \frac{1}{\lambda}\int_0^\lambda a_{i,e}\, di. \tag{29}$$

As in the single-signal case, the coordination target is the expert average $\bar{a}_e$; the only change relative to Section 2 is that agents now condition their posterior beliefs on $(x_{i,\tau}, y^L, y^H)$ rather than $(x_{i,\tau}, y)$.

As in the single-signal case, the coordination motive causes experts to overweight public signals relative to their Bayesian precision weights: the denominator is $\alpha_L + \alpha_H + 1 - r$ rather than $\alpha_L + \alpha_H + 1$, so the private signal receives strictly less weight than its precision would warrant, and the expert's strategy deviates from the fundamental.

The presence of the expert precision (normalized to unity) in the public's weight expressions confirms that the public's optimal action depends on experts' information, reflecting the public's incentive to coordinate with experts. As in the single-signal case, one can verify that $k_{L,e} < k_{L,p}$ and $k_{H,e} < k_{H,p}$: the public overweights both public signals relative to experts, consistent with its lower private precision $\beta < 1$.

Comparing the weights assigned to the two public signals, both groups place greater weight on the high-precision signal: $k_{L,e} < k_{H,e}$ and $k_{L,p} < k_{H,p}$. The ratio of weights is identical across groups:

$$\frac{k_{H,e}}{k_{L,e}} = \frac{k_{H,p}}{k_{L,p}} = \frac{\alpha_H}{\alpha_L}. \tag{30}$$

Although both groups overreact to public information in aggregate due to the coordination motive, the two public signals are weighted proportionally to their precisions within the composite public signal. The asymmetry between experts and the public operates on total reliance on public information, not on how that reliance is allocated between the two signals.

4.3 Welfare Gains from Differentiated Communication

Suppose the social planner has already determined the optimal single precision α^* from Proposition 1. In the differentiated-signal regime, the high-quality signal retains this precision, $\alpha_H^* = \alpha^*$, and the social planner chooses an additional low-quality precision $\alpha_L \in [0, \alpha_H^*)$. The social welfare expression $W_2(\alpha_L, \alpha_H^*)$ is derived in Appendix B. The same three-component decomposition as in equation (18) applies, with the composite precision $\alpha_L + \alpha_H^*$ replacing α and the cost term becoming $(1-\lambda)(q_L\alpha_L + q_H\alpha_H^*)$; the full expression is given in Appendix B. We differentiate W_2 with respect to α_L and evaluate

the two boundary cases.

Releasing a near-vacuous second signal. Consider the limit $\alpha_L \to 0$. Since $q_L = q_H \cdot (\alpha_L/\alpha_H^*) < q_H$, a signal of vanishing precision is also trivially cheap to interpret. Evaluating the welfare function and its derivative at $\alpha_L = 0$ gives

$$W_2(0, \alpha_H^*) = W_1(\alpha^*), \qquad \left.\frac{\partial W_2}{\partial \alpha_L}\right|_{\alpha_L=0} = \left.\frac{\partial W_1}{\partial \alpha}\right|_{\alpha=\alpha^*} + (1-\lambda)q_H > 0. \tag{31}$$

The second equality exploits the first-order optimality condition $\partial W_1/\partial\alpha|_{\alpha=\alpha^*} = 0$; the strict positivity then follows from $q_H > 0$. This shows that adding a marginally positive-precision low-quality signal strictly raises welfare relative to the single-signal optimum: even a minimally informative but maximally accessible second signal reduces the effective per-unit cost of expanding the public's information set.

Releasing a second signal identical to the first. At the other extreme, set $\alpha_L = \alpha_H^* = \alpha^*$. Then the two signals are observationally equivalent and

$$W_2(\alpha_H^*, \alpha_H^*) = W_1(2\alpha^*) < W_1(\alpha^*), \qquad \left.\frac{\partial W_2}{\partial \alpha_L}\right|_{\alpha_L=\alpha_H^*} = \left.\frac{\partial W_1}{\partial \alpha}\right|_{\alpha=2\alpha^*} - (1-\lambda)q_H < 0. \tag{32}$$

Duplicating the original signal does not reduce comprehension difficulty and doubles interpretation cost, so welfare falls. Equation (32) provides a formal counterpart to the intuition in Blinder et al. (2022) that communication targeted at the general public should be meaningfully distinct from expert-level communication, not merely a repetition of it.

Existence of an interior optimum. Combining equations (31) and (32), $\partial W_2/\partial\alpha_L$ changes sign over the interval $(0, \alpha_H^*)$: it is strictly positive at $\alpha_L = 0$ and strictly negative at $\alpha_L = \alpha_H^*$. By continuity, there exists an interior $\alpha_L^* \in (0, \alpha_H^*)$ at which W_2 is maximized. For $\alpha_L \in (0, \bar{\alpha}_L)$, where $\bar{\alpha}_L$ is the crossing point at which $W_2(\alpha_L, \alpha_H^*) = W_1(\alpha^*)$, the differentiated-signal regime is strictly welfare-improving relative to the single-signal optimum. For $\alpha_L \in (\bar{\alpha}_L, \alpha_H^*)$ the two signals are too similar to yield a comprehension benefit and the social planner is better off releasing only one. Figure 3 depicts this relationship numerically under the benchmark calibration.

Welfare under the differentiated-signal policy strictly exceeds welfare under the optimal single-signal policy (see Figure 3), and the welfare-maximizing low-precision signal has precision substantially below α_H^*, confirming that the signal for the general public should be genuinely simpler than the expert-level signal.

A second signal helps only if it is *genuinely different* from the first. When α_L is close to α_H^*, releasing a second signal amounts to repeating the first in only slightly plainer

terms: this fails to reduce interpretation difficulty appreciably and still incurs non-trivial cost. A substantially simplified signal, one that replaces precise technical language with plain prose, charts, or summary statistics, cuts interpretation costs sharply while still conveying directional information about the fundamentals. The planner therefore sets α_L^* strictly below α_H^*, exploiting the precision gap to redirect the cost saving toward improved comprehension among the general public. This result complements the findings of Cornand and Heinemann (2008) and Baeriswyl and Cornand (2014) on the optimality of partial transparency: whereas those papers establish that limiting the *audience* of a public signal can be optimal when coordination is strong, the present model establishes that differentiating the *content* of signals across audience types is welfare-improving even when all signals are publicly broadcast.

Proposition 3 (Welfare gain from differentiated communication)**.** *Under Assumption 2, when the economy contains two types of heterogeneous agents, the social planner can strictly improve on the single-signal optimum $W_1(\alpha^*)$ by additionally releasing a low-precision public signal. There exists a threshold $\bar{\alpha}_L \in (0, \alpha_H^*)$ such that $W_2(\alpha_L, \alpha_H^*) > W_1(\alpha^*)$ for every $\alpha_L \in (0, \bar{\alpha}_L)$. On this interval W_2 is strictly concave in α_L, so the maximum is attained at a unique interior point $\alpha_L^* \in (0, \bar{\alpha}_L)$. Releasing two signals of identical precision, $\alpha_L = \alpha_H^*$, is welfare-inferior.*

Proof. See Appendix B. □

4.4 Comparative Statics of Signal Differentiation

We examine how the optimal differentiation ratio α_L^*/α_H^* responds to the key parameters.

Public precision β. Figure 4a plots α_L^*/α_H^* against $\beta \in (0, 1]$. The ratio is increasing in β: as the public's own private information deteriorates (smaller β), the social planner should issue a *simpler* complementary signal, i.e. one whose precision is further below α_H^*. When β is small, the general public gains the most from accessible information, so the marginal welfare return to simplicity is higher and the optimal α_L^* shifts downward relative to α_H^*.

Expert share λ. Figure 4b shows that α_L^*/α_H^* is decreasing in λ. When the expert share is large, the high-quality signal y^H is optimally set at a high precision (from Proposition 2), which makes it correspondingly harder for the remaining general public to interpret. The social planner therefore needs a lower-precision complementary signal to bridge the comprehension gap.

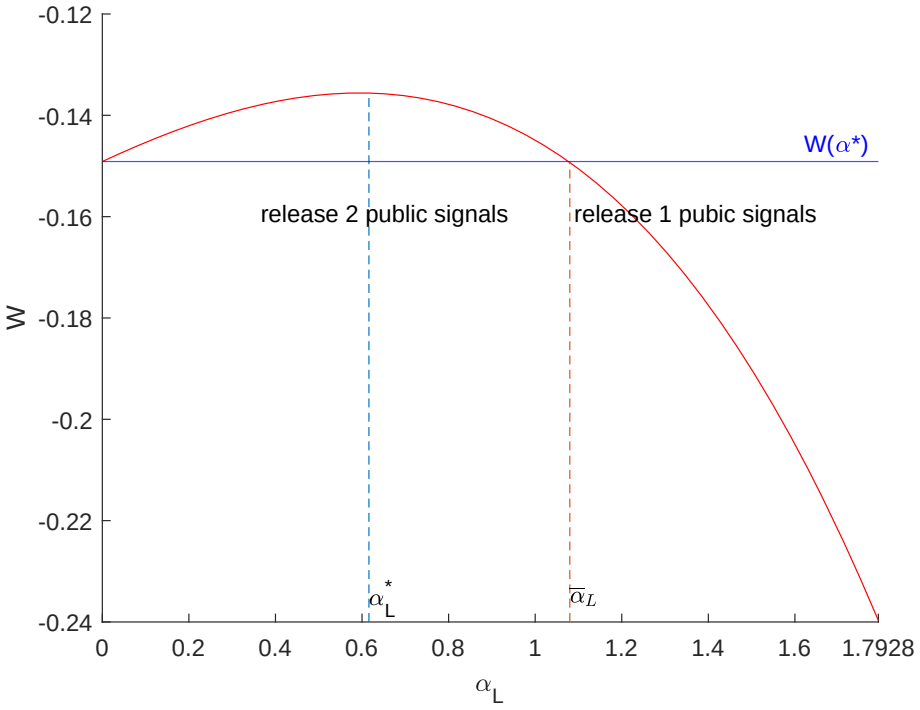


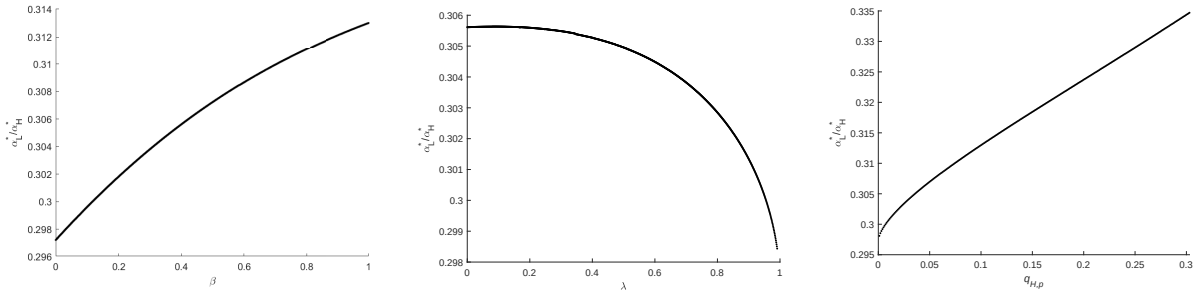
Figure 3: Social welfare under the differentiated-signal regime.
Note: The red curve shows $W_2(\alpha_L, \alpha_H^*)$ as a function of α_L; the blue line shows $W_1(\alpha^*)$. Parameters: $r = 0.85$, $\lambda = 0.005$, $\beta = 0.4$, $q = 0.04$.

Marginal interpretation cost q_H. Figure 4c plots α_L^*/α_H^* against $q_H \in (0, \bar{q})$. The ratio is increasing in q_H. This may look counterintuitive, since a higher cost of interpreting technical information calls for a low-precision signal closer in precision to the high-precision one; the mechanism is in fact an envelope effect. A higher q_H reduces α_H^* itself (Proposition 2), which shrinks the feasible precision gap and forces the planner to concentrate informational content in the simpler channel.

Proposition 4 (Optimal degree of signal differentiation)**.** *The social planner widens the precision gap between the two signals, that is, decreases α_L^*/α_H^*, when (i) the precision of the public's private signal β falls, (ii) the expert share λ rises, or (iii) the marginal interpretation cost q_H falls.*

Proof. See Appendix B. □

Proposition 4 yields a practical rule: the precision gap between the expert signal and the public signal should be larger when audience capabilities are more dispersed, that is, when the public has weaker private information, when the expert sector is larger, or when the marginal cost of simplification is low. Propositions 3 and 4 together show that the optimal degree of differentiation responds in a clean way to audience characteristics, which makes the framework directly amenable to calibration.



(a) α_L^*/α_H^* vs. β (b) α_L^*/α_H^* vs. λ (c) α_L^*/α_H^* vs. q_H

Figure 4: Comparative statics of the optimal precision gap α_L^*/α_H^* in the differentiated-signal regime.
Note: Panel (a): sensitivity to β. Panel (b): sensitivity to λ. Panel (c): sensitivity to q_H. Benchmark: $r = 0.85$, $\lambda = 0.005$, $\beta = 0.4$, $q = 0.04$.

5 Selective Interpretation

The analysis so far assumes that the general public processes every public signal released by the central bank. In practice, households and small investors have limited attention and choose which signals to read. Following Arato and Nakamura (2023), we endogenize the public's interpretation decision. Given the menu of signals chosen by the planner, the general public selects the subset that maximizes expected utility and pays interpretation costs only for the signals it actually processes. Experts continue to observe all signals because they face no interpretation cost. We solve the planner's problem by backward induction.

Under the single-signal policy of Section 3, this extension adds no new choice for the public. The non-trivial case is the differentiated-signal policy of Section 4. Let $E(u_{i,p}^{all})$ denote expected utility when the public reads both signals, $E(u_{i,p}^{L})$ when it reads only the low-precision signal, and $E(u_{i,p}^{H})$ when it reads only the high-precision signal. The public chooses the option that delivers the highest expected utility.

The derivations in Appendix C show that the public may strictly prefer to read both signals, so the benchmark differentiated-signal analysis remains internally consistent. They also show that when the precision gap $\alpha_H - \alpha_L$ is small, the incremental benefit of the high-precision signal can fall below its interpretation cost. In that case the public rationally reads only the low-precision signal. When the gap is larger, the low-precision signal is cheap enough to decode that reading both signals again becomes optimal.

Proposition 5 (Selective interpretation under differentiated communication)**.** *When the public's interpretation strategy is endogenous, there exist parameter configurations under which the general public rationally chooses to interpret only one of the two available signals. When the precision gap $\alpha_H - \alpha_L$ is small, the marginal information gain from the high-precision signal falls below its interpretation cost, and the public optimally reads only y^L. The social planner's optimal policy still involves releasing two signals with differentiated*

precision, and the optimal low-precision signal satisfies $\tilde{\alpha}_L^* > \alpha_L^*$*, where* α_L^* *is the benchmark optimum of Section 4.*

The social welfare expressions under the three interpretation scenarios are given in Appendix C. Under the benchmark parameterization, the public rationally switches from reading both signals to reading only the simpler one when the precision gap falls below a critical threshold α_L^{crit}. The planner responds by raising the optimal low-precision signal above the benchmark α_L^*, steering the public toward the simpler channel. The resulting welfare level exceeds that of the benchmark differentiated-signal equilibrium, so attention steering is itself a source of welfare gain (see Figure 5).

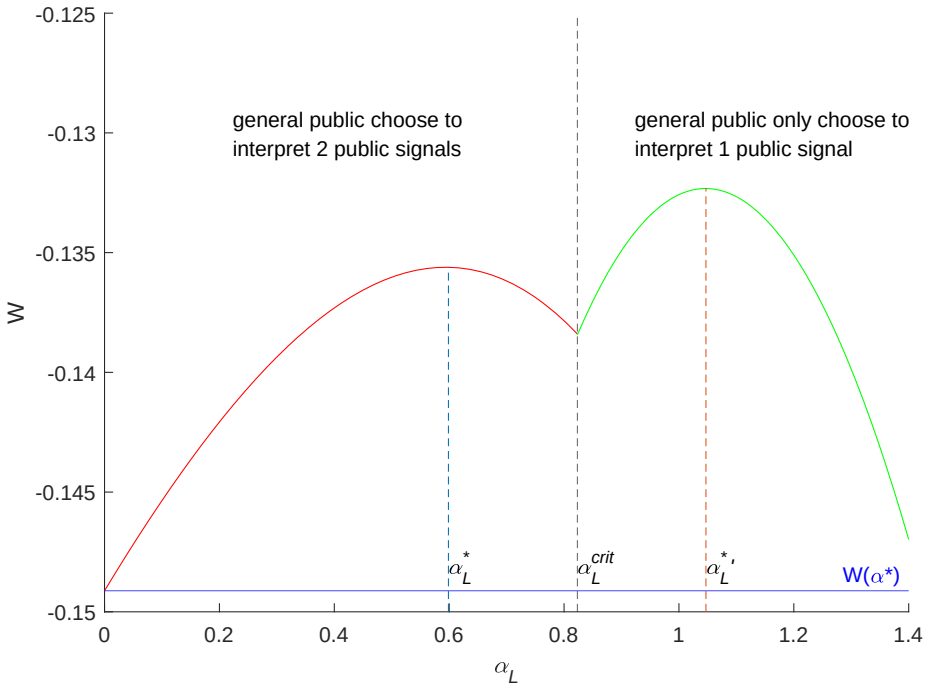


Figure 5: Social planner's strategy with endogenous information interpretation. Note: The figure compares welfare under different interpretation strategies when the general public can selectively process signals. Parameters: $r = 0.85$, $\lambda = 0.005$, $\beta = 0.4$, $q = 0.04$.

6 Conclusion

This paper studies how a central bank should design public communication when experts and the general public differ in private information, interpretation costs, and coordination motives. Under a single public signal, a precision effect, an interpretation-cost effect, and a crowding-out effect jointly make welfare non-monotone in precision, so the optimal precision is strictly interior. The central bank can strictly improve on this single-signal optimum by issuing differentiated signals: the high-precision signal preserves expert-side

accuracy while the low-precision signal reduces the cognitive burden on the general public, and these complementary roles generate a welfare gain that does not depend on restricting access to either signal.

Comparative statics tie the optimal precision gap directly to audience heterogeneity: a less-informed public, a larger expert sector, or lower simplification costs all call for greater differentiation, so the communication architecture should widen as household financial literacy falls. When the public chooses endogenously which signals to read, the planner raises the precision of the low-precision signal to steer attention toward the cheaper channel, which reinforces rather than undermines the case for differentiated communication.

Policy implications. The results support a clearly differentiated communication architecture for central banks. Technical publications such as policy reports, forward guidance, and fan charts serve the expert audience; plain-language summaries, visual releases, and concrete numerical examples serve the general public. The precision of the plain-language release should match the comprehension capacity of households and retail investors. When the central bank can also influence salience (for example, by actively promoting the simpler release through its official channels), the model predicts an additional welfare gain from attention steering.

Limitations and future research. Three limitations invite extension. First, interpretation costs here are exogenous and linear in signal precision. Richer specifications, such as costs that are convex, depend on distance from prior beliefs, or interact with aggregate uncertainty shocks, could generate additional non-monotonicities and help account for the observed oscillation of central banks between technical and accessible registers. Second, the model abstracts from strategic communication motives. Embedding the planner's precision choice in a repeated game with reputational costs for deviating from expected transparency would let the framework address the persistence of communication norms across policy cycles. Third, the beauty-contest payoff structure fits asset-pricing and wage-setting contexts well but requires adaptation for inflation-expectation settings, where the social cost of miscoordination is linked to the aggregate gap between expected and realized inflation rather than to cross-sectional action dispersion. Replacing the coordination term with a New-Keynesian Phillips-curve mechanism would bring the framework closer to the literature on communication and inflation anchoring (Coibion et al., 2023; D'Acunto et al., 2022).

References

Allen, Franklin, Stephen Morris, and Hyun Song Shin (2006) "Beauty Contests and Iterated Expectations in Asset Markets," *Review of Financial Studies*, 19 (3), 719–752.

Andre, Peter, Carlo Pizzinelli, Christopher Roth, and Johannes Wohlfart (2022) "Subjective Models of the Macroeconomy: Evidence from Experts and Representative Samples," *Review of Economic Studies*, 89 (6), 2958–2991.

Angeletos, George-Marios and Chen Lian (2018) "Forward Guidance without Common Knowledge," *American Economic Review*, 108 (9), 2477–2512.

Angeletos, George-Marios and Alessandro Pavan (2007) "Efficient Use of Information and Social Value of Information," *Econometrica*, 75 (4), 1103–1142.

Arato, Hiroki, Takeo Hori, and Tomoya Nakamura (2021) "Endogenous Information Acquisition and the Partial Announcement Policy," *Information Economics and Policy*, 55, 100898.

Arato, Hiroki and Tomoya Nakamura (2023) "Welfare Effects of Partial Publicity and Partial Transparency in Endogenous Information Acquisition," Working Paper.

Bacchetta, Philippe and Eric van Wincoop (2006) "Can Information Heterogeneity Explain the Exchange Rate Determination Puzzle?" *American Economic Review*, 96 (3), 552–576.

Baeriswyl, Romain, Kene Boun My, and Camille Cornand (2021) "Double Overreaction in Beauty Contests with Information Acquisition: Theory and Experiment," *Journal of Monetary Economics*, 118, 432–445.

Baeriswyl, Romain and Camille Cornand (2014) "Reducing Overreaction to Central Banks' Disclosures: Theory and Experiment," *Journal of the European Economic Association*, 12 (4), 1087–1126.

Bholat, David, Nida Broughton, Janna Ter Meer, and Eryk Walczak (2019) "Enhancing Central Bank Communications Using Simple and Relatable Information," *Journal of Monetary Economics*, 108, 1–15.

Blinder, Alan S. (2009) "Talking about Monetary Policy: The Virtues (and Vices?) of Central Bank Communication," Technical Report274, Bank for International Settlements, `https://www.bis.org/publ/work274.htm`.

——— (2018) "Through a Crystal Ball Darkly: The Future of Monetary Policy Communication," *AEA Papers and Proceedings*, 108, 567–571.

Blinder, Alan S., Michael Ehrmann, Jakob de Haan, and David-Jan Jansen (2022) "Central Bank Communication with the General Public: Promise or False Hope?", Working Paper.

Boun My, Kene, Camille Cornand, and R. D. S. Ferreira (2021) "Public Information and the Concern for Coordination," *Journal of Behavioral and Experimental Economics*, 93, 101710.

Chan, Siu Lung (2021) "The Social Value of Public Information When Not Everyone Is Privately Informed," Working Paper.

Coenen, Günter, Michael Ehrmann, Gaetano Gaballo, Peter Hoffmann, Anton Nakov, Stefano Nardelli, Eric Persson, and Georg Strasser (2017) "Communication of Monetary Policy in Unconventional Times," working paper, European Central Bank.

Coibion, Olivier and Yuriy Gorodnichenko (2015) "Information Rigidity and the Expectations Formation Process: A Simple Framework and New Facts," *American Economic Review*, 105 (8), 2644–2678.

Coibion, Olivier, Yuriy Gorodnichenko, and Michael Weber (2022) "Monetary Policy Communications and Their Effects on Household Inflation Expectations," *Journal of Political Economy*, 130 (6), 1537–1584.

——— (2023) "Forward Guidance and Household Expectations," *Journal of the European Economic Association*, 21 (5), 2131–2171.

Colombo, Luca and Gianluca Femminis (2008) "The Social Value of Public Information with Costly Information Acquisition," *Economics Letters*, 100, 196–199.

——— (2014) "Optimal Policy Intervention, Constrained Obfuscation and the Social Value of Public Information," *Economics Letters*, 123, 224–226.

Cornand, Camille and Frank Heinemann (2008) "Optimal Degree of Public Information Dissemination," *Economic Journal*, 118 (528), 718–742.

D'Acunto, Francesco, Daniel Hoang, Tarun Ramadorai, and Michael Weber (2022) "Managing Households' Expectations with Unconventional Policies," *Review of Financial Studies*, 35 (4), 1597–1642.

Draghi, Mario (2014) "Central Bank Communication," Speech, European Central Bank PR Manager of the Year Award.

Ehrmann, Michael and Alena Wabitsch (2022) "Central Bank Communication with Non-Experts—A Road to Nowhere?" *Journal of Monetary Economics*, 127, 69–85.

Fullwood, James (2016) "A Cat, a Hat and a Simple Measure of Gobbledygook: How Readable Is Your Writing?", Bank Underground blog post.

Grossman, Sanford J. and Joseph E. Stiglitz (1980) "On the Impossibility of Informationally Efficient Markets," *American Economic Review*, 70 (3), 393–408.

Grout, Paul A., Sébastien Mitraille, and Silvia Sonderegger (2015) "The Costs and Benefits of Coordinating with a Different Group," *Journal of Economic Theory*, 160, 517–535.

Haldane, Andrew (2017) "A Little More Conversation, a Little Less Action," Speech, Bank of England.

Haldane, Andrew, Alistair Macaulay, and Michael McMahon (2020) "The 3 E's of Central Bank Communication with the Public," Discussion Paper14265, Centre for Economic Policy Research.

Haldane, Andrew and Michael McMahon (2018) "Central Bank Communications and the General Public," *AEA Papers and Proceedings*, 108, 578–583.

Hansen, Stephen, Michael McMahon, and Matthew Tong (2019) "The Long-Run Information Effect of Central Bank Communication," *Journal of Monetary Economics*, 108, 185–202.

Hoffmann, Mathias, Emanuel Moench, Lora Pavlova, and Guido Schultefrankenfeld (2022) "Would Households Understand Average Inflation Targeting?" *Journal of Monetary Economics*, 129, S52–S66.

Indjejikian, Raffi J. (1991) "The Impact of Costly Information Interpretation on Firm Disclosure Decisions," *Journal of Accounting Research*, 29 (2), 277–301.

Istrefi, Klodiana (2019) "Comment on: Enhancing Central Bank Communications Using Simple and Relatable Information," *Journal of Monetary Economics*, 108, 16–20.

Kozicki, Sharon and Jessica Vardy (2017) "Communicating Uncertainty in Monetary Policy," Staff Working Paper 14, Bank of Canada.

Morris, Stephen and Hyun Song Shin (2002) "Social Value of Public Information," *American Economic Review*, 92 (5), 1521–1534.

——— (2007) "Optimal Communication," *Journal of the European Economic Association*, 5 (2–3), 594–602.

Morris, Stephen, Hyun Song Shin, and Hui Tong (2006) "Social Value of Public Information: Morris and Shin (2002) Is Actually Pro-Transparency, Not Con: Reply," *American Economic Review*, 96 (1), 453–455.

Myatt, David P. and Chris Wallace (2014) "Central Bank Communication Design in a Lucas–Phelps Economy," *Journal of Monetary Economics*, 63, 64–79.

Orhun, Eda (2019) "Disclosure Decision in an Entry Game with Costly Information Interpretation," *Global Economy Journal*, 19 (1), 1950005.

Powell, Jerome H. (2018) "Transcript of Chairman Powell's Press Conference, June 13," Federal Reserve Board.

Svensson, Lars E. O. (2006) "Social Value of Public Information: Comment: Morris and Shin (2002) Is Actually Pro-Transparency, Not Con," *American Economic Review*, 96 (1), 448–452.

Ui, Takashi (2014) "The Social Value of Public Information with Convex Costs of Information Acquisition," *Economics Letters*, 125 (2), 249–252.

van der Cruijsen, Carin, David-Jan Jansen, and Jakob de Haan (2015) "How Much Does the Public Know about the ECB's Monetary Policy? Evidence from a Survey of Dutch Households," *International Journal of Central Banking*, 11, 169–218.

Verrecchia, Robert E. (1982) "Information Acquisition in a Noisy Rational Expectations Economy," *Econometrica*, 50 (6), 1415–1430.

Woodford, Michael (2003) "Imperfect Common Knowledge and the Effects of Monetary Policy," in Philippe Aghion, Roman Frydman, Joseph Stiglitz, and Michael Woodford eds. *Knowledge, Information, and Expectations in Modern Macroeconomics: In Honor of Edmund S. Phelps*, Princeton University Press.

——— (2005) "Central-Bank Communication and Policy Effectiveness," in *The Greenspan Era: Lessons for the Future*, 399–474: Federal Reserve Bank of Kansas City.

Appendix A Proofs for the Single-Signal Model

Proof of Lemma 1

Proof. The individual utility maximization problem for agent i of type τ is

$$\max_{a_{i,\tau}} \; u_{i,\tau} = -(1-r)(a_{i,\tau}-\theta)^2 - r(L_{i,\tau}-\bar{L}) - C_\tau(\alpha), \tag{33}$$

where

$$L_{i,\tau} = \frac{1}{\lambda}\int_0^\lambda (a_{i,\tau}-a_{j,e})^2\,dj, \qquad \bar{L} = \int_0^\lambda L_{j,e}\,dj + \int_\lambda^1 L_{j,p}\,dj. \tag{34}$$

Because agent i has measure zero in $[0,1]$, her own action $a_{i,\tau}$ influences $\bar{L}$ only through $\bar{a}_e$, and only on a null set; $\bar{L}$ is therefore taken as independent of $a_{i,\tau}$ when differentiating utility (see also Angeletos and Pavan, 2007). The first-order condition with respect to $a_{i,\tau}$ is

$$\frac{\partial u_{i,\tau}}{\partial a_{i,\tau}} = -2(1-r)(a_{i,\tau}-\theta) - 2r\left(a_{i,\tau} - \frac{1}{\lambda}\int_0^{\lambda} a_{j,e}\,dj\right) = 0, \tag{35}$$

which yields the optimal best-response

$$a_{i,\tau} = (1-r)E_{i,\tau}(\theta) + rE_{i,\tau}(\bar{a}_e), \qquad \bar{a}_e = \frac{1}{\lambda}\int_0^{\lambda} a_{i,e}\,di. \tag{36}$$

Expert equilibrium. Conjecture a linear strategy for experts: $a^1_{i,e} = (1-k_e)x_i + k_e y$, so that $\bar{a}^1_e = (1-k_e)\theta + k_e y$. The expert's posterior is (with expert precision normalized to unity, $\beta_e = 1$)

$$E_{i,e}(\theta \mid x_i, y) = \frac{x_i + \alpha y}{\alpha + 1}. \tag{37}$$

Substituting (37) and the conjectured average action into (36):

$$a^1_{i,e} = (1-rk_e)\frac{x_i + \alpha y}{\alpha+1} + rk_e y = \frac{(1-rk_e)}{\alpha+1}\,x_i + \frac{\alpha + rk_e}{\alpha+1}\,y. \tag{38}$$

Matching coefficients with the conjecture gives $k_e = (\alpha + rk_e)/(\alpha+1)$, hence

$$k_e = \frac{\alpha}{\alpha + (1-r)}. \tag{39}$$

The unique linear equilibrium action for the expert is therefore

$$a^1_{i,e} = \frac{(1-r)}{\alpha+(1-r)}\,x_i + \frac{\alpha}{\alpha+(1-r)}\,y. \tag{40}$$

Higher-order expectations and the fixed point. Setting $\mu = 1/(\alpha+1)$, iterating the best-response (36) and expanding via the tower property yields

$$a_{i,e} = (1-r)\sum_{k=0}^{\infty} r^k E_{i,e}\big(\bar{E}^k_e(\theta)\big)\,. \tag{41}$$

We prove by induction that for any k,

$$\bar{E}^k_e(\theta) = (1-\mu^k)y + \mu^k\theta, \qquad E_{i,e}\big(\bar{E}^k_e(\theta)\big) = (1-\mu^{k+1})y + \mu^{k+1}x_{i,e}.$$

Base case ($k=1$): $\bar{E}_e(\theta) = (\alpha y+\theta)/(\alpha+1) = (1-\mu)y+\mu\theta$. $E_{i,e}(\bar{E}_e(\theta)) = (1-\mu^2)y+\mu^2 x_{i,e}$ by direct computation.

Inductive step: assume the claim holds for $k-1$. Then

$$E_{i,e}\big(\bar{E}_e^{k-1}(\theta)\big) = (1-\mu^k)y + \mu^k x_{i,e},$$

so $\bar{E}_e^k(\theta) = (1-\mu^k)y + \mu^k\theta$, and

$$E_{i,e}\big(\bar{E}_e^{k}(\theta)\big) = (1-\mu^k)y + \mu^k\left(\frac{\alpha y + x_{i,e}}{\alpha+1}\right) = (1-\mu^{k+1})y + \mu^{k+1}x_{i,e}.$$

This completes the induction. Substituting into (41):

$$a_{i,e} = (1-r)\sum_{k=0}^{\infty} r^k\big[(1-\mu^{k+1})y + \mu^{k+1}x_{i,e}\big] = \left[1 - \frac{\mu(1-r)}{1-r\mu}\right]y + \frac{\mu(1-r)}{1-r\mu}\,x_{i,e} = \frac{\alpha y + (1-r)x_{i,e}}{\alpha+(1-r)}, \tag{42}$$

which is exactly (40), confirming uniqueness of the linear equilibrium.

Public equilibrium. The public's conjecture is $a^1_{i,p} = (1-k_p)x_{i,p} + k_p y$. The public's posterior is

$$E_{i,p}(\theta \mid x_i, y) = \frac{\alpha y + \beta x_{i,p}}{\alpha+\beta}. \tag{43}$$

Substituting into (36) and using (39):

$$a^1_{i,p} = (1-rk_e)\frac{\alpha y + \beta x_{i,p}}{\alpha+\beta} + rk_e y = \frac{(1-rk_e)\beta}{\alpha+\beta}\,x_{i,p} + \frac{\alpha + rk_e\beta}{\alpha+\beta}\,y, \tag{44}$$

so

$$k_p = \frac{\alpha + rk_e\beta}{\alpha+\beta}, \qquad \frac{\partial k_p}{\partial k_e} = \frac{r\beta}{\alpha+\beta} > 0. \tag{45}$$

Substituting (39) into (44):

$$a^1_{i,p} = \frac{(1-r)(\alpha+1)\beta}{(\alpha+\beta)[\alpha+(1-r)]}\,x_{i,p} + \frac{[\alpha+(1-r)+r\beta]\,\alpha}{(\alpha+\beta)[\alpha+(1-r)]}\,y. \tag{46}$$

One verifies that $k_p > k_e$ because $\beta < \beta_e = 1$:

$$k_p - k_e = \frac{\alpha(1-r)(1-\beta)}{[\alpha+(1-r)](\alpha+\beta)} > 0. \tag{47}$$

The public, having a less precise private signal, overweights the public signal relative to experts. □

Derivation of Social Welfare and the Decomposition

Social welfare equals the cross-sectional average of individual utilities:

$$W_1 = E\left(\int_0^1 u_i \, di\right) = E\left[-(1-r)\int_0^1 (a_i^1 - \theta)^2 \, di\right] - (1-\lambda)q\alpha. \tag{48}$$

Expanding by agent type and using $E(v_{i,e}^2) = 1$, $E(v_{i,p}^2) = 1/\beta$, $E(\eta^2) = 1/\alpha$:

$$W_1 = -\lambda(1-r)\frac{\alpha + (1-r)^2}{[\alpha + (1-r)]^2} - (1-\lambda)(1-r)\frac{\alpha + r^2 k_e^2 \beta}{\alpha(\alpha+\beta)} - (1-\lambda)q\alpha. \tag{49}$$

For transparency, we decompose W_1 into three effects.

Precision effect. Absent coordination and interpretation costs, the welfare gain from precision is the reduction in expected squared error, which is the sum of the expert and public precision terms in (49):

$$W_{\text{precision}} = -(1-r)\left[\lambda\frac{1}{\alpha+1} + (1-\lambda)\frac{1}{\alpha+\beta}\right]. \tag{50}$$

Information cost effect. The cost of information provision is $q\alpha$ per agent, so the total cost is $(1-\lambda)q\alpha$ since only the public pays. This reduces welfare by

$$W_{\text{cost}} = -(1-\lambda)q\alpha. \tag{51}$$

Crowding-out effect. The difference between the full welfare and the sum of the precision and cost effects is the crowding-out term:

$$W_{\text{crowd-out}} = -(1-r)\left[\lambda\frac{r^2\alpha}{[\alpha+(1-r)]^2(\alpha+1)} + (1-\lambda)\frac{r^2\beta\alpha}{[\alpha+(1-r)]^2(\alpha+\beta)}\right] \leq 0. \tag{52}$$

The crowding-out term is zero when $r = 0$ and strictly negative for $r > 0$: agents over-react to the public signal for coordination reasons, suppressing their more precise private signals and pushing the optimal precision below the first-best level.

Proof of Lemma 2

Proof. The first-order condition for α can be written as

$$\frac{\partial W_1}{\partial \alpha} = F(\alpha) - (1-\lambda)q, \tag{53}$$

where $F(\alpha)$ collects the precision and crowding-out contributions:

$$F(\alpha) = (1-r)\left[\frac{\lambda}{(\alpha+1)^2}\big(1+g_1(\alpha)\big) + \frac{(1-\lambda)}{(\alpha+\beta)^2}\big(1+g_2(\alpha)\big)\right], \tag{54}$$

with

$$g_1(\alpha) = \frac{r^2\big(2\alpha^2+\alpha-(1-r)\big)}{[\alpha+(1-r)]^3}, \qquad g_2(\alpha) = \frac{r^2\beta\big(2\alpha^2+\beta\alpha-(1-r)\beta\big)}{[\alpha+(1-r)]^3}. \tag{55}$$

Part (i). When $r \le 1/2$, we have $g_1(0) = -r^2/(1-r)^2 \ge -1$, so $1+g_1(\alpha) \ge 0$ for all $\alpha \ge 0$. Since $g_2(0) = -r^2\beta^2/(1-r)^2 > g_1(0)\cdot\beta^2 > -1$, both terms in $F(\alpha)$ are non-negative, whence $F(\alpha) > 0$ for all $\alpha > 0$.

Part (ii). When $r > 1/2$, define $f_1(\alpha) = 2\alpha^2+\alpha-(1-r)$ and $f_2(\alpha) = 2\alpha^2+\beta\alpha-(1-r)\beta$. The root of f_1 is $\bar{\alpha}_1 = \big(\sqrt{1+8(1-r)}-1\big)/4$ and g_1 inherits the sign of f_1. One verifies that $g_1'(\alpha) = 2r^2(\alpha+1)(2(1-r)-\alpha)/[\alpha+(1-r)]^4$, so g_1 is hump-shaped on $(0, 2(1-r))$.

Because $1+g_1(0) < 0$ and $1+g_1(\alpha) \to 1$ as $\alpha \to \infty$, the function $1+g_1$ crosses zero exactly once at some $\hat{\alpha}_1$. Similarly, $1+g_2$ has at most one zero $\hat{\alpha}_2 \le \hat{\alpha}_1$. On the interval $(\hat{\alpha}_2, \hat{\alpha}_1)$, $1+g_1 < 0 < 1+g_2$.

To show that F has a unique zero on $(\hat{\alpha}_2, \hat{\alpha}_1)$, define $h_1(\alpha) \equiv (1+g_1(\alpha))/(1+g_2(\alpha))$ and $h_2(\alpha) \equiv -(1-\lambda)(\alpha+1)^2/[\lambda(\alpha+\beta)^2]$; the equation $F(\alpha) = 0$ reduces to $h_1(\alpha) = h_2(\alpha)$. Direct differentiation shows that $h_1'(\alpha) > 0$ and $h_1''(\alpha) < 0$ on this interval, while $h_2'(\alpha) > 0$ and $h_2''(\alpha) < 0$ with h_2 strictly dominating h_1 near $\hat{\alpha}_1$. We check the sign of $h_1 - h_2$ by calculating it for a fine grid of parameter values. It turns out the signed difference $h_1 - h_2$ is strictly concave and changes sign exactly once, which delivers the claimed single zero $\hat{\alpha}$. □

Proof of Proposition 1

Proof. Differentiating (49) with respect to α and collecting terms yields

$$\begin{aligned}\frac{\partial W_1}{\partial \alpha} = &-\lambda(1-r)\frac{-\alpha+(1-r)(2r-1)}{[\alpha+(1-r)]^3} \\ &-(1-\lambda)\frac{1-r}{[\alpha+(1-r)]^3}\left\{\frac{[\alpha+(1-r)][(\alpha+\beta)^2-[(1-r)-\beta]^2+r^2\beta^2]}{(\alpha+\beta)^2}\right. \\ &\left.-\frac{2(1-r)^2(\alpha+1)^2}{\alpha+\beta} - 2\alpha\left[2r(1-r)\frac{\alpha+1}{\alpha+\beta}+r^2\right]\right\} - (1-\lambda)q.\end{aligned} \tag{56}$$

Taking the limit as $\alpha \to \infty$:

$$\lim_{\alpha\to\infty}\frac{\partial W_1}{\partial \alpha} = -(1-\lambda)q < 0. \tag{57}$$

Hence for sufficiently large α, additional precision reduces welfare. By Lemma 2, when $r > 1/2$ the net marginal benefit $F(\alpha)$ is negative near $\alpha = 0$, reaches a unique zero at $\hat{\alpha}$, and is positive for $\alpha > \hat{\alpha}$. For the full derivative $\partial W_1/\partial\alpha = F(\alpha) - (1-\lambda)q$, the cost term shifts F downward by the constant $(1-\lambda)q$; under Assumption 2 ($q < \bar{q}$), the shifted function is strictly positive on a non-empty interval (α_1, α_2). Since the derivative is negative both near zero and as $\alpha \to \infty$, the intermediate value theorem applied on (α_1, α_2) and (α_2, ∞) guarantees at least one interior α^* where $\partial W_1/\partial\alpha = 0$. Strict concavity of W_1 at this critical point (verified analytically under $r > 1/2$) implies uniqueness. □

Proof of Proposition 2

Proof. The optimal precision α^* is implicitly defined by the first-order condition

$$F(\alpha, \Theta) \equiv \frac{\partial W_1}{\partial \alpha} = 0, \tag{58}$$

where $\Theta = (\beta, \lambda, q)$. Since α^* is a strict maximum, $F_\alpha(\alpha^*, \Theta) < 0$. By the implicit function theorem, for any parameter $\theta \in \Theta$:

$$\operatorname{sign}\left(\frac{\partial \alpha^*}{\partial \theta}\right) = \operatorname{sign}(F_\theta).$$

(i) $\partial\alpha^*/\partial\beta < 0$. Applying the normalization (expert precision equal to unity):

$$F_\beta = -\frac{2(1-\lambda)(1-r)^2\big[(1+\alpha)^3 + r^2(1+\alpha\beta) - r(2-\alpha)(1+\alpha)^2\big]}{[\alpha + (1-r)]^3(\alpha+\beta)^3}. \tag{59}$$

The numerator is positive for $r \in (1/2, 1)$ and $\alpha > 0$, so $F_\beta < 0$.

(ii) $\partial\alpha^*/\partial\lambda > 0$. Direct differentiation yields

$$F_\lambda = \frac{(1-r)\big(1 - r(3-2r) + \alpha\big)}{[\alpha + (1-r)]^3(1-\lambda)}. \tag{60}$$

At the optimum $\alpha^* > r(3-2r) - 1$ (which is verified by evaluating F at the critical point; see below), hence $F_\lambda > 0$.

To confirm that $\alpha^* > r(3-2r) - 1$, evaluate F at $\alpha = r(3-2r) - 1$:

$$F\big(r(3-2r) - 1\big) = \frac{[4(1-r)r + \beta](1-\beta)(1-\lambda)}{4r[1 - (3-2r)r - \beta]^2} > 0. \tag{61}$$

Since $F_\alpha < 0$ at α^* and $F(\alpha^*) = 0$, the positivity of F at $r(3-2r) - 1$ implies $\alpha^* > r(3-2r) - 1$.

(iii) $\partial\alpha^*/\partial q < 0$. Since $F_q = -(1-\lambda) < 0$, the result follows immediately. □

Appendix B Proofs for the Dual-Signal Model

This appendix proves the results of Section 4. Throughout, two public signals y^L (low-precision, α_L) and y^H (high-precision, α_H, $\alpha_H > \alpha_L$) are released simultaneously. Interpretation costs are q_L and q_H respectively.

Derivation of Equilibrium Actions under the Dual-Signal Regime

Expert strategy. Conjecture $a_{i,e}^{L,H} = (1 - k_{L,e} - k_{H,e})x_{i,e} + k_{L,e}y^L + k_{H,e}y^H$, so that

$$\bar{a}_e^{L,H} = (1 - k_{L,e} - k_{H,e})\theta + k_{L,e}y^L + k_{H,e}y^H. \tag{62}$$

The expert's posterior combines the two public signals into a composite $Z = \lambda_1 y^L + (1 - \lambda_1)y^H$ obtained by minimizing $\mathrm{Var}(\lambda_1 y^L + (1 - \lambda_1)y^H - \theta) = \lambda_1^2/\alpha_L + (1 - \lambda_1)^2/\alpha_H$, yielding $\lambda_1 = \alpha_L/(\alpha_L + \alpha_H)$ and composite precision

$$\alpha_Z = \alpha_L + \alpha_H. \tag{63}$$

Hence

$$E_{i,e}(\theta \mid x_{i,e}, y^L, y^H) = \frac{\alpha_L y^L + \alpha_H y^H + x_{i,e}}{\alpha_L + \alpha_H + 1}. \tag{64}$$

Substituting into the best-response and matching coefficients:

$$a_{i,e}^{L,H} = \frac{(1-r)}{\alpha_L + \alpha_H + (1-r)}\, x_{i,e} + \frac{\alpha_L}{\alpha_L + \alpha_H + (1-r)}\, y^L + \frac{\alpha_H}{\alpha_L + \alpha_H + (1-r)}\, y^H. \tag{65}$$

As in the single-signal case, the expert over-weights public signals relative to precision weights to achieve coordination.

Public strategy. By the same composite-signal argument the public forms

$$E_{i,p}(\theta \mid x_{i,p}, y^L, y^H) = \frac{\alpha_L y^L + \alpha_H y^H + \beta x_{i,p}}{\alpha_L + \alpha_H + \beta}. \tag{66}$$

Substituting into the best-response and using (65):

$$a_{i,p}^{L,H} = \frac{1}{\alpha_H + \alpha_L + (1-r)} \Bigg\{ (1-r)\frac{\alpha_H + \alpha_L + 1}{\alpha_H + \alpha_L + \beta}\, \beta\, x_{i,p} + \frac{\alpha_H + \alpha_L + (1-r) + r\beta}{\alpha_H + \alpha_L + \beta} \big(\alpha_L y^L + \alpha_H y^H\big) \Bigg\}. \tag{67}$$

Public overreacts relative to experts. $\beta < 1$ implies $(\alpha_L+\alpha_H+1)/(\alpha_L+\alpha_H+\beta) > 1$, so $k_{L,p} > k_{L,e}$ and $k_{H,p} > k_{H,e}$. Both agent types overweight the high-precision signal: $k_{L,e} < k_{H,e}$ and $k_{L,p} < k_{H,p}$ follow directly from $\alpha_L < \alpha_H$.

Derivation of Social Welfare under the Dual-Signal Regime

$$
\begin{aligned}
W_2 = &-\lambda(1-r)\frac{\alpha_L+\alpha_H+(1-r)^2}{[\alpha_L+\alpha_H+(1-r)]^2} \\
&-(1-\lambda)(1-r)\frac{\left[(1-r)\dfrac{\alpha_L+\alpha_H+1}{\alpha_L+\alpha_H+\beta}+r\right]^2(\alpha_L+\alpha_H)+(1-r)^2\left(\dfrac{\alpha_L+\alpha_H+1}{\alpha_L+\alpha_H+\beta}\right)^2\beta}{[\alpha_L+\alpha_H+(1-r)]^2} \\
&-(1-\lambda)(q_L\alpha_L+q_H\alpha_H).
\end{aligned} \tag{68}
$$

The partial derivative with respect to α_L has the same functional form as $\partial W_1/\partial\alpha$ (Equation (56)) with α replaced by $\alpha_L+\alpha_H$ and q replaced by q_L.

Proof of Proposition 3

Proof. We verify two key inequalities.

First inequality: $W_2(0,\alpha_H^*) = W_1(\alpha^*)$. When $\alpha_L = 0$ the dual-signal model collapses to the single-signal model with precision α_H^*. If the social planner sets α_H^* optimally in the single-signal problem, then $W_2(0,\alpha_H^*) = W_1(\alpha^*)$ by construction.

Second inequality: $\left.\dfrac{\partial W_2(\alpha_L,\alpha_H^*)}{\partial\alpha_L}\right|_{\alpha_L=0} > 0.$

$$
\begin{aligned}
\left.\frac{\partial W_2}{\partial\alpha_L}\right|_{\alpha_L=0} = &-\lambda\left[(1-r)\frac{-\alpha_H^*+(1-r)(2r-1)}{[\alpha_H^*+(1-r)]^3}\right] \\
&-\frac{(1-\lambda)(1-r)}{[\alpha_H^*+(1-r)]^3}\left\{\frac{(\alpha_H^*+(1-r))[(\alpha_H^*+\beta)^2-[(1-r)-\beta]^2+r^2\beta^2]}{(\alpha_H^*+\beta)^2}\right. \\
&\left.-\frac{2(1-r)^2(\alpha_H^*+1)^2}{\alpha_H^*+\beta}-2\alpha_H^*\left[2r(1-r)\frac{\alpha_H^*+1}{\alpha_H^*+\beta}+r^2\right]\right\}.
\end{aligned} \tag{69}
$$

Recognising that the above equals $\partial W_1/\partial\alpha|_{\alpha=\alpha^*} + (1-\lambda)q = 0 + (1-\lambda)q > 0$ (since α^* satisfies the first-order condition $\partial W_1/\partial\alpha = 0$), the derivative is strictly positive for any $q > 0$. By continuity there exists $\Delta\alpha_L > 0$ such that $W_2(\alpha_L,\alpha_H^*) > W_1(\alpha^*)$ for all $\alpha_L \in (0,\Delta\alpha_L)$. □

Proof that Two Identical Signals are Welfare-Inferior to One

When $\alpha_L = \alpha_H = \alpha^*$, the composite precision is $2\alpha^*$ and the total interpretation cost is $(q_L + q_H)\alpha^* = 2q\alpha^*$ under symmetric marginal costs, so

$$W_2(\alpha^*, \alpha^*) = W_1(2\alpha^*) < W_1(\alpha^*), \tag{70}$$

the last inequality holding because α^* uniquely maximizes W_1.

The derivative at this boundary satisfies

$$\left.\frac{\partial W_2}{\partial \alpha_L}\right|_{\alpha_L = \alpha_H^*} = \left.\frac{\partial W_1}{\partial \alpha}\right|_{\alpha = 2\alpha^*} - (1-\lambda)q < 0, \tag{71}$$

since W_1 is decreasing beyond its interior maximum α^*, so $\partial W_1/\partial\alpha|_{\alpha=2\alpha^*} < 0$, and the subtracted term is positive. The social planner therefore strictly prefers to reduce α_L below α_H^*.

Proof of Proposition 4

Proof. We work in the sequential framework where $A \equiv \alpha_H^* = \alpha^*$ is the single-signal optimum from Proposition 1 and α_L^* maximizes $W_2(\alpha_L, A)$.

Step 1: Structural equation for the ratio. Because W_2 depends on α_L through the composite precision $S = \alpha_L + A$ (in the precision and crowding-out components) and through the quadratic cost $q\alpha_L^2/A$ (arising from $q_L = q\alpha_L/A$ via (25)), the first-order condition $\partial W_2/\partial\alpha_L = 0$ is

$$F(\alpha_L^* + A) = \frac{2(1-\lambda)q\,\alpha_L^*}{A}, \tag{72}$$

where $F(\cdot)$ is the function defined in (54). Dividing (72) by the single-signal first-order condition $F(A) = (1-\lambda)q$ yields the *ratio equation*

$$\frac{F\big((1+\rho)A;\beta,\lambda\big)}{F(A;\beta,\lambda)} = 2\rho, \qquad \rho \;\equiv\; \frac{\alpha_L^*}{A}. \tag{73}$$

Since F is strictly decreasing on $[A,\infty)$ we have $F((1+\rho)A)/F(A) < 1$, whence $\rho < \frac{1}{2}$. The second-order condition $F'(S)A < 2F(A)$ holds automatically ($F'(S) < 0$ and $F(A) > 0$), guaranteeing that (73) has a unique solution $\rho \in (0, \frac{1}{2})$.

Step 2: Elasticity characterization. Define the *elasticity of* F on its decreasing branch:

$$\varepsilon(\alpha) \;\equiv\; -\frac{\alpha\, F'(\alpha)}{F(\alpha)} \;>\; 0, \qquad \alpha \geq A.$$

Taking logarithms of (73) and totally differentiating with respect to a parameter $\theta \in \{\beta, \lambda, q\}$ (noting that $A = \alpha_H^*(\theta)$ and $S = (1+\rho)A$) gives, after rearranging,

$$\frac{d\rho}{d\theta} = \frac{\big[\varepsilon(A) - \varepsilon(S)\big]\dfrac{1}{A}\dfrac{dA}{d\theta} \;+\; \big[L_\theta(S) - L_\theta(A)\big]}{\dfrac{1}{\rho} + \dfrac{\varepsilon(S)}{1+\rho}}, \tag{74}$$

where $L_\theta(\alpha) = \partial \log F(\alpha; \beta, \lambda)/\partial\theta$ is the semi-elasticity of F with respect to θ. The denominator is strictly positive.

Step 3: Monotonicity properties of F. From the explicit expressions (54)–(55) three properties hold on $[A, \infty)$:

(P1) $\varepsilon(\alpha)$ is strictly increasing, approaching 2 from below as $\alpha \to \infty$.

(P2) $L_\beta(\alpha) = \partial \log F/\partial\beta < 0$ is strictly increasing (i.e. $|L_\beta|$ is decreasing).

(P3) $L_\lambda(\alpha) = \partial \log F/\partial\lambda$ is non-increasing.

Verification of (P1). For large α the corrections g_1, g_2 in (55) are $O(1/\alpha)$, so $F(\alpha) \sim (1-r)/\alpha^2$ and $\varepsilon(\alpha) \approx 2\alpha/(\alpha + c)$ for an effective shift $c > 0$ determined by β and r. Each component $(1+g_j)/(\alpha + a_j)^2$ (with $a_1 = 1$, $a_2 = \beta$) has individually increasing elasticity $2\alpha/(\alpha + a_j)$; since F is a positive linear combination of these components, ε inherits the monotonicity.

Verification of (P2). $F_\beta < 0$ by (59). The sensitivity concentrates in the public term $(1-\lambda)(1+g_2)/(\alpha+\beta)^2$, whose log-derivative with respect to β satisfies $|L_\beta(\alpha)| = O(1/\alpha)$ and is monotonically diminishing.

Verification of (P3). $F_\lambda = (1-r)\big[(1+g_1)/(\alpha+1)^2 - (1+g_2)/(\alpha+\beta)^2\big]$ is positive at the optimum (verified in the proof of Proposition 2). The ratio $L_\lambda = F_\lambda/F$ converges to a constant as $\alpha \to \infty$, so L_λ is non-increasing on the relevant interval.

Step 4: Sign determination. *(iii)* $d\rho/dq > 0$. F does not depend on q, so $L_q \equiv 0$. From Proposition 2(iii), $dA/dq < 0$. By (P1), $\varepsilon(A) < \varepsilon(S)$, so the numerator of (74) is

$$\underbrace{[\varepsilon(A) - \varepsilon(S)]}_{<\,0} \cdot \underbrace{\frac{1}{A}\frac{dA}{dq}}_{<\,0} \;>\; 0.$$

A positive numerator divided by a positive denominator gives $d\rho/dq > 0$: the ratio α_L^*/α_H^* decreases when q falls, widening the precision gap.

(i) $d\rho/d\beta > 0$. From Proposition 2(i), $dA/d\beta < 0$. The numerator of (74) consists of two terms:

$$\underbrace{[\varepsilon(A) - \varepsilon(S)]}_{<0} \cdot \underbrace{\frac{1}{A}\frac{dA}{d\beta}}_{<0} + \underbrace{[L_\beta(S) - L_\beta(A)]}_{>0 \text{ by (P2)}}.$$

Both terms are positive, so $d\rho/d\beta > 0$: the ratio decreases when β falls, widening the precision gap.

(ii) $d\rho/d\lambda < 0$. From Proposition 2(ii), $dA/d\lambda > 0$. The numerator is

$$\underbrace{[\varepsilon(A) - \varepsilon(S)]}_{<0} \cdot \underbrace{\frac{1}{A}\frac{dA}{d\lambda}}_{>0} + \underbrace{[L_\lambda(S) - L_\lambda(A)]}_{\leq 0 \text{ by (P3)}}.$$

Both terms are non-positive (the first is strictly negative), so $d\rho/d\lambda < 0$: the precision gap widens as the expert share rises. □

Appendix C Proofs for Endogenous Information Interpretation

This appendix derives the individual utility expressions used in Section 5. Recall the three public-information interpretation strategies available to the general public when the social planner operates under the dual-signal regime: interpret both (*all*), interpret only the low-precision signal (L), or interpret only the high-precision signal (H).

Strategy *all*: Interpreting Both Signals

When the public interprets both signals, their strategy follows (67):

$$a_{i,p}^{all} = \frac{(1-r)\dfrac{\alpha_H + \alpha_L + 1}{\alpha_H + \alpha_L + \beta}\,\beta \cdot x_{i,p}}{\alpha_H + \alpha_L + (1-r)} + \frac{\left[(1-r)\dfrac{\alpha_H + \alpha_L + 1}{\alpha_H + \alpha_L + \beta} + r\right](\alpha_L y^L + \alpha_H y^H)}{\alpha_H + \alpha_L + (1-r)}. \quad (75)$$

The resulting individual utility is

$$u_{i,p}^{all} = -(1-r)\frac{\left[(1-r)\dfrac{\alpha_L+\alpha_H+1}{\alpha_L+\alpha_H+\beta}+r\right]^2(\alpha_L+\alpha_H)+(1-r)^2\left(\dfrac{\alpha_L+\alpha_H+1}{\alpha_L+\alpha_H+\beta}\right)^2\beta}{[\alpha_L+\alpha_H+(1-r)]^2} \tag{76}$$

$$-\frac{r\lambda}{[\alpha_L+\alpha_H+(1-r)]^2}\Bigg[-(1-r)^2\beta+(1-r)^2\left(\frac{\alpha_L+\alpha_H+1}{\alpha_L+\alpha_H+\beta}\right)^2\beta + \left[(1-r)\frac{\alpha_L+\alpha_H+1}{\alpha_L+\alpha_H+\beta}+r-1\right]^2(\alpha_L+\alpha_H)\Bigg]-q_L\alpha_L-q_H\alpha_H.$$

Strategy L: Interpreting Only the Low-Precision Signal

When the public reads only y^L, the expert's average action still depends on both signals, but the public's posterior is based on $(x_{i,p}, y^L)$ alone. Define

$$a_{i,p}^L = \frac{\dfrac{(1-r)(\alpha_L+1)+\alpha_H}{\alpha_L+\beta}\beta\cdot x_{i,p}}{\alpha_L+\alpha_H+(1-r)} + \frac{\left[\dfrac{(1-r)(\alpha_L+1)+\alpha_H}{\alpha_L+\beta}+r\right]\alpha_L\cdot y^L}{\alpha_L+\alpha_H+(1-r)}. \tag{77}$$

The resulting individual utility is

$$u_{i,p}^L = -(1-r)\frac{\left[\dfrac{\alpha_H+(1-r)(\alpha_L+1)}{\alpha_L+\beta}+r\right]^2\alpha_L+\left(\dfrac{\alpha_H+(1-r)(\alpha_L+1)}{\alpha_L+\beta}\right)^2\beta}{[\alpha_L+\alpha_H+(1-r)]^2} \tag{78}$$

$$-q_L\alpha_L-\frac{r\lambda}{[\alpha_L+\alpha_H+(1-r)]^2}\Bigg\{-(1-r)^2+\left(\frac{\alpha_H+(1-r)(\alpha_L+1)}{\alpha_L+\beta}\right)^2\beta + \left[\frac{\alpha_H+(1-r)(\alpha_L+1)}{\alpha_L+\beta}+r-1\right]^2\alpha_L+\alpha_H\Bigg\}.$$

Strategy H: Interpreting Only the High-Precision Signal

Symmetrically, when the public reads only y^H:

$$a_{i,p}^H = \frac{\dfrac{(1-r)(\alpha_H+1)+\alpha_L}{\alpha_H+\beta}\beta\cdot x_{i,p}}{\alpha_L+\alpha_H+(1-r)} + \frac{\left[\dfrac{(1-r)(\alpha_H+1)+\alpha_L}{\alpha_H+\beta}+r\right]\alpha_H\cdot y^H}{\alpha_L+\alpha_H+(1-r)}. \tag{79}$$

The resulting individual utility is

$$u_{i,p}^{H} = -(1-r)\frac{\left[\dfrac{\alpha_L+(1-r)(\alpha_H+1)}{\alpha_H+\beta}+r\right]^2\alpha_H+\left(\dfrac{\alpha_L+(1-r)(\alpha_H+1)}{\alpha_H+\beta}\right)^2\beta}{[\alpha_L+\alpha_H+(1-r)]^2} \tag{80}$$
$$-q_H\alpha_H-\frac{r\lambda}{[\alpha_L+\alpha_H+(1-r)]^2}\Bigg\{-(1-r)^2+\left(\frac{\alpha_L+(1-r)(\alpha_H+1)}{\alpha_H+\beta}\right)^2\beta$$
$$+\left[\frac{\alpha_L+(1-r)(\alpha_H+1)}{\alpha_H+\beta}+r-1\right]^2\alpha_H+\alpha_L\Bigg\}.$$

Equilibrium Interpretation Strategy

The public chooses argmax$\{E(u_{i,p}^{all}),\ E(u_{i,p}^{L}),\ E(u_{i,p}^{H})\}$.

Case 1 (both signals always interpreted): Under the parameterisation $r = 0.85$, $\lambda = 0.35$, $\beta = 0.4$, $q = 0.04$, direct computation confirms $E(u_{i,p}^{all}) > E(u_{i,p}^{L})$ and $E(u_{i,p}^{all}) > E(u_{i,p}^{H})$ for all (α_L, α_H) in the feasible set, as illustrated in Figure A1.

Case 2 (selective attention): Under the benchmark parameterisation of Section 4 ($\lambda = 0.005$), there exists a threshold $\tilde{\alpha}_L$ such that the public interprets both signals when $\alpha_L < \tilde{\alpha}_L$ (wide precision gap, low decoding cost for y^L) and only y^L when $\alpha_L > \tilde{\alpha}_L$ (narrow gap, insufficient incremental benefit from y^H), as illustrated in Figure A2.

The social welfare under each scenario follows directly from substituting the relevant action into the aggregate welfare expression. The welfare expressions for the L-only and H-only cases are

$$W_L = -\lambda(1-r)\frac{\alpha_L+\alpha_H+(1-r)^2}{[\alpha_L+\alpha_H+(1-r)]^2}-(1-\lambda)q_L\alpha_L \tag{81}$$
$$-(1-\lambda)(1-r)\frac{\left[\dfrac{\alpha_H+(1-r)(\alpha_L+1)}{\alpha_L+\beta}+r\right]^2\alpha_L+\left(\dfrac{\alpha_H+(1-r)(\alpha_L+1)}{\alpha_L+\beta}\right)^2\beta}{[\alpha_L+\alpha_H+(1-r)]^2},$$

$$W_H = -\lambda(1-r)\frac{\alpha_L+\alpha_H+(1-r)^2}{[\alpha_L+\alpha_H+(1-r)]^2}-(1-\lambda)q_H\alpha_H \tag{82}$$
$$-(1-\lambda)(1-r)\frac{\left[\dfrac{\alpha_L+(1-r)(\alpha_H+1)}{\alpha_H+\beta}+r\right]^2\alpha_H+\left(\dfrac{\alpha_L+(1-r)(\alpha_H+1)}{\alpha_H+\beta}\right)^2\beta}{[\alpha_L+\alpha_H+(1-r)]^2}.$$

Proof of Proposition 5

Proof. Since experts face no interpretation cost, they always read both signals and their action is fixed across all three scenarios. Comparing interpretation strategies therefore

reduces to comparing the public's per-capita welfare (after subtracting the common expert loss term). We analyze each case in turn.

Case 1: Public reads only y^L. Using the action (77), the public's per-capita welfare contribution is

$$W_L = M(\alpha_L) - q\,\frac{\alpha_L^2}{\alpha_H^*}, \tag{83}$$

where $M(\alpha_L) = -(1-r)\dfrac{1+\dfrac{r^2\beta\,\alpha_L}{[\alpha_L+\alpha_H^*+(1-r)]^2}}{\alpha_L+\beta}$. Define $\eta(\alpha_L) = 1+r^2\beta\,\alpha_L/[\alpha_L+\alpha_H^*+(1-r)]^2$. Then

$$\eta'(\alpha_L) = \frac{r^2\beta\big(\alpha_H^*+(1-r)-\alpha_L\big)}{[\alpha_L+\alpha_H^*+(1-r)]^3},$$

and

$$\frac{\partial W_L}{\partial \alpha_L} = (1-r)\frac{\alpha_L^3+(3F+2r^2\beta)\alpha_L^2+(3F^2+r^2\beta^2)\alpha_L+F^3-r^2\beta^2F}{(\alpha_L+\beta)^2[\alpha_L+\alpha_H^*+(1-r)]^3} - \frac{2q\,\alpha_L}{\alpha_H^*}, \tag{84}$$

where $F = \alpha_H^* + (1-r) > 0$. The second derivative satisfies $\partial^2 W_L/\partial\alpha_L^2 < 0$, so W_L is strictly concave and first increases then decreases.

Case 2: Public reads only y^H. Differentiating (82) with respect to α_L:

$$\frac{\partial W_H}{\partial \alpha_L} = \frac{2(1-r)r^2\alpha_H^*\beta}{(\alpha_H^*+\beta)[\alpha_L+\alpha_H^*+(1-r)]^3} > 0, \tag{85}$$

and $\partial^2 W_H/\partial\alpha_L^2 < 0$. Hence W_H is strictly increasing and concave in α_L.

Case 3: Public reads both signals. Using the action (75), the welfare W_{all} is computed in (76). The derivative satisfies

$$\frac{\partial W_{all}}{\partial \alpha_L} = (1-r)\frac{[\alpha_L+\alpha_H^*+(1-r)]^3+r^2\beta\big[(\alpha_L+\alpha_H^*)(2(\alpha_L+\alpha_H^*)+\beta)-(1-r)\beta\big]}{[\alpha_L+\alpha_H^*+(1-r)]^3(\alpha_L+\alpha_H^*+\beta)^2} - \frac{2q\,\alpha_L}{\alpha_H^*}, \tag{86}$$

and W_{all} is concave and first increases then decreases under Assumption 2.

Welfare ranking. We establish the ranking at the boundaries:

At $\alpha_L = \alpha_H^*$: $W_L = W_H$, and by Proposition 3, $W_L = W_H > W_1(\alpha_H^*)$.

At $\alpha_L = 0$: $W_L(0) = -(1-r)/\beta$, while $W_H(0) = W_{all}(0) = W_1(\alpha^*) > W_L(0)$.

First-order comparison at $\alpha_L = 0$:

$$\left.\frac{\partial W_{all}}{\partial \alpha_L}\right|_{\alpha_L=0} - \left.\frac{\partial W_H}{\partial \alpha_L}\right|_{\alpha_L=0} = \frac{(1-r)[\alpha_H^*+(1-r)][\alpha_H^*+(1-r)-r\beta^2][\alpha_H^*+(1-r)+r\beta^2]}{[\alpha_H^*+(1-r)]^3(\alpha_H^*+\beta)^2} > 0. \tag{87}$$

Crossing-point analysis. Since W_H is increasing and concave, and W_L is concave with $W_L(\alpha_H^*) = W_H(\alpha_H^*)$ but $W_H(0) > W_L(0)$, these two functions cross exactly once at some $\alpha_L^1 \in (0, \alpha_H^*)$. By the same concavity argument, W_{all} and W_L cross at a unique

$\alpha_L^{crit} \in (0, \alpha_H^*)$, and W_{all} and W_H cross at $\alpha_L' \in (0, \alpha_H^*)$. The derivative comparison (87) ensures $0 < \alpha_L^1 < \alpha_L^{crit} < \alpha_L' < \alpha_H^*$, yielding:

- For $\alpha_L \in (0, \alpha_L^{crit})$: $W_{all} > W_H > W_L$, so the public reads both signals.
- For $\alpha_L \in (\alpha_L^{crit}, \alpha_H^*)$: $W_L > W_H > W_{all}$, so the public reads only y^L.

Hence there exist parameter configurations under which the general public rationally chooses to interpret only one signal. As stated in the body text, the social planner's optimal α_L^* may lie in either region, and when it falls in $(\alpha_L^{crit}, \alpha_H^*)$ the effective low-precision signal receives all of the public's attention. □

Appendix D Numerical Analysis of Correlated Public Signals

We relax the assumption of conditional independence between the two public signals. The social planner releases

$$\begin{pmatrix} y^L \\ y^H \end{pmatrix} = \begin{pmatrix} \theta \\ \theta \end{pmatrix} + \begin{pmatrix} \eta^L \\ \eta^H \end{pmatrix}, \qquad \begin{pmatrix} \eta^L \\ \eta^H \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1/\alpha_L & \rho_{LH}/\sqrt{\alpha_L \alpha_H} \\ \rho_{LH}/\sqrt{\alpha_L \alpha_H} & 1/\alpha_H \end{pmatrix} \right), \tag{88}$$

where $\rho_{LH} \in (0, 1)$ is the correlation between the two noise terms.

Expert Strategy under Correlated Signals

Conjecture $a_{i,e}^{corr} = (1 - f_{L,e} - f_{H,e})x_{i,e} + f_{L,e}y^L + f_{H,e}y^H$, so $\bar{a}_e^{corr} = (1 - f_{L,e} - f_{H,e})\theta + f_{L,e}y^L + f_{H,e}y^H$.

The composite public signal $Q = \lambda_2 y^L + (1 - \lambda_2)y^H$ that minimizes $\mathrm{Var}(Q - \theta)$ over the correlated bivariate normal has optimal weight

$$\lambda_2 = \frac{\alpha_L - \rho_{LH}\sqrt{\alpha_L \alpha_H}}{\alpha_L + \alpha_H - 2\rho_{LH}\sqrt{\alpha_L \alpha_H}}, \tag{89}$$

and composite precision

$$\alpha_Q = \frac{\alpha_L + \alpha_H - 2\rho_{LH}\sqrt{\alpha_L \alpha_H}}{1 - \rho_{LH}^2}. \tag{90}$$

Note that $\alpha_Q = \alpha_L + \alpha_H$ when $\rho_{LH} = 0$, recovering (63). For $\rho_{LH} > 0$, $\alpha_Q < \alpha_L + \alpha_H$: correlation reduces the informational content of the composite signal.

The expert's posterior is

$$E_{i,e}(\theta \mid x_{i,e}, y^L, y^H) = (1 - \tilde{\lambda}_2)x_{i,e} + \tilde{\lambda}_2 Q, \qquad \tilde{\lambda}_2 = \frac{\alpha_Q}{\alpha_Q + 1}, \tag{91}$$

and the equilibrium strategy is

$$a_{i,e}^{corr} = \frac{(1-r)}{\alpha_Q + (1-r)}\, x_{i,e} + \frac{(\alpha_L - \rho_{LH}\sqrt{\alpha_L \alpha_H})y^L + (\alpha_H - \rho_{LH}\sqrt{\alpha_L \alpha_H})y^H}{[\alpha_Q + (1-r)](1-\rho_{LH}^2)}. \tag{92}$$

Public Strategy under Correlated Signals

By the same projection argument, the public's posterior uses the same α_Q:

$$\begin{aligned} a_{i,p}^{corr} &= (1-r)\frac{\alpha_Q + 1}{\alpha_Q + \beta}\frac{\beta}{\alpha_Q + (1-r)}\, x_{i,p} \\ &\quad + \left[(1-r)\frac{\alpha_Q + 1}{\alpha_Q + \beta} + r\right] \frac{(\alpha_L - \rho_{LH}\sqrt{\alpha_L \alpha_H})y^L + (\alpha_H - \rho_{LH}\sqrt{\alpha_L \alpha_H})y^H}{[\alpha_Q + (1-r)](1-\rho_{LH}^2)}. \end{aligned} \tag{93}$$

Social Welfare under Correlated Signals

$$\begin{aligned} W_2^{corr} &= -\lambda(1-r)\left[\frac{(1-r)^2}{[\alpha_Q + (1-r)]^2} + \frac{\alpha_L + \alpha_H - 2\rho_{LH}\sqrt{\alpha_L \alpha_H}}{[\alpha_Q + (1-r)]^2(1-\rho_{LH}^2)}\right] \\ &\quad - (1-\lambda)(1-r)\left[\frac{\left[(1-r)\dfrac{\alpha_Q + 1}{\alpha_Q + \beta}\right]^2 \beta}{[\alpha_Q + (1-r)]^2} + \frac{\left[(1-r)\dfrac{\alpha_Q + 1}{\alpha_Q + \beta} + r\right]^2 (\alpha_L + \alpha_H - 2\rho_{LH}\sqrt{\alpha_L \alpha_H})}{[\alpha_Q + (1-r)]^2(1-\rho_{LH}^2)}\right] \\ &\quad - (1-\lambda)(q_L \alpha_L + q_H \alpha_H). \end{aligned} \tag{94}$$

Effect of Signal Correlation on Optimal Policy

Equation (94) has the same structural form as (68), with $\alpha_L + \alpha_H$ replaced by $\alpha_Q < \alpha_L + \alpha_H$ whenever $\rho_{LH} > 0$. Since α_Q decreases in ρ_{LH}, stronger correlation effectively reduces the precision of the composite public signal, tightening the constraint facing the social planner.

Numerically (Figures 6, parameterisation $r = 0.85$, $\lambda = 0.005$, $\beta = 0.4$, $q = 0.04$), two results emerge. First, for weak correlation ($\rho_{LH} = 0.15$), the dual-signal policy continues to dominate the single-signal benchmark: there exists an interior $\alpha_L^* > 0$ that maximizes W_2^{corr}. Second, as ρ_{LH} increases, the optimal α_L^* falls toward zero (Figure 7): when the two signals are highly correlated their joint informational content barely exceeds that of a single signal, so the interpretation cost of the second signal cannot be justified. The policy implication is that the central bank should ensure its simpler communication differs not only in precision but also in *informational content* from its technical releases, for instance by basing simpler releases on different indicators or framing them around different aspects of the economic outlook.

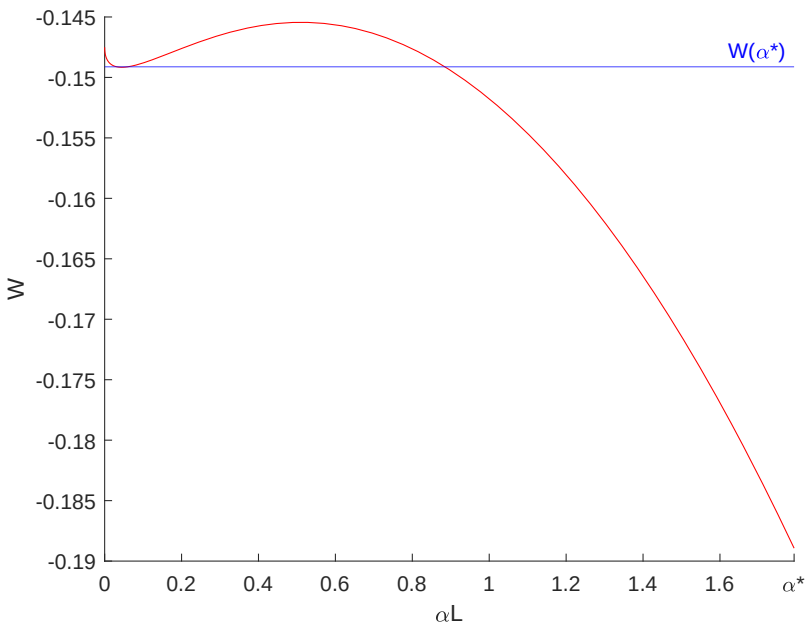


Figure 6: Social-planner strategy with two weakly correlated public signals. Note: W_2^{corr} as a function of α_L under weak correlation $\rho_{LH} = 0.15$. Parameters: $r = 0.85$, $\beta = 0.4$, $\lambda = 0.005$, $q = 0.04$.

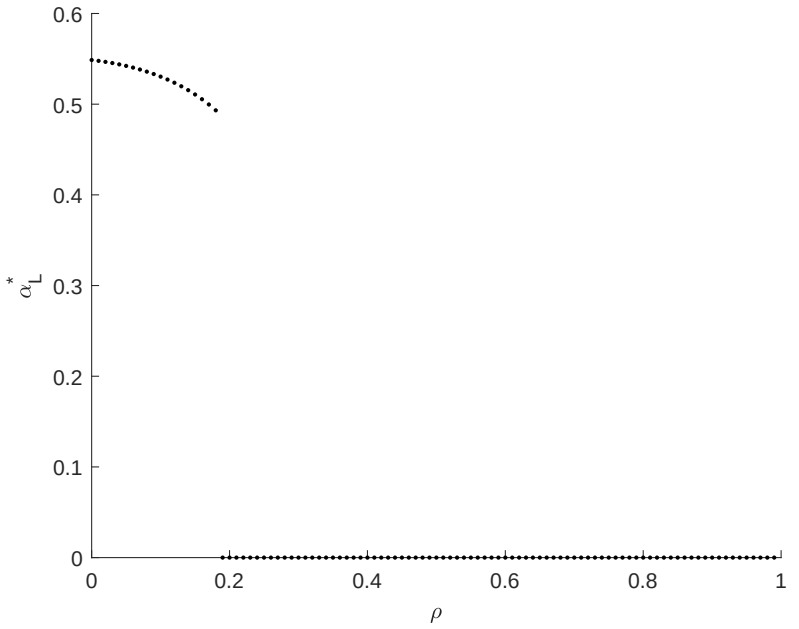


Figure 7: Optimal α_L^* as a function of signal correlation ρ_{LH}. Note: The figure traces the welfare-maximizing low-precision signal α_L^* as the correlation ρ_{LH} between the two public signals varies. Parameters: $r = 0.85$, $\beta = 0.4$, $\lambda = 0.005$, $q = 0.04$.